

IMPORTANT 12-MARK PYQ - MOST REPEATED ★

EXAM ALERT
12 MARKS
VVI QUESTIONS

Q1. Explain CPU scheduling in Operating System. Discuss its criteria, algorithms with suitable example.

Ans.

1. INTRODUCTION :

CPU scheduling is the process of selecting a process from the ready queue and allocating CPU to it. The objective is to maximize CPU utilization and to provide good performance.

2. CPU SCHEDULING CRITERIA :

- ① CPU Utilization - Maximize
- ② Throughput - Maximize
- ③ Turnaround Time - Minimize
- ④ Waiting Time - Minimize
- ⑤ Response Time - Minimize
- ⑥ Fairness - Avoid starvation

3. CPU SCHEDULING ALGORITHMS :

- (A) **FCFS (First Come First Serve)**
Process with earliest arrival time is executed first. It is non-preemptive.
- (B) **SJF (Shortest Job First)**
Process with shortest CPU burst time is executed first. It is non-preemptive.
- (C) **SRTF (Shortest Remaining Time First)**
Preemptive version of SJF. Process with shortest remaining time is executed.
- (D) **Priority Scheduling**
Process with highest priority is executed first. Preemptive / Non-preemptive.
- (E) **Round Robin (RR)**
Each process gets CPU for a fixed time quantum in circular order. It is preemptive.

Memory Trick

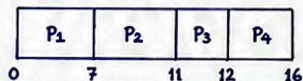
For Criteria

U - Utilization
T - Throughput
T - Turnaround
W - Waiting
R - Response
F - Fairness
(**UTTWRF**)

4. EXAMPLE :

Processes	Arrival Time	Burst Time
P ₁	0	7
P ₂	2	4
P ₃	4	1
P ₄	5	4

(i) Gantt Chart - FCFS

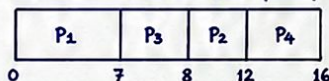


WT (Waiting Time) =

P₁=0, P₂=7-2=5, P₃=11-4=7, P₄=12-5=7

Avg WT = (0+5+7+7)/4 = 4.75

(ii) Gantt Chart - SJF (Non-preemptive)



WT = P₁=0, P₃=7-4=3, P₂=8-2=6, P₄=12-5=7

Avg WT = (0+3+6+7)/4 = 4.00

5. CONCLUSION : CPU scheduling is essential for efficient utilization of CPU. Different algorithms are used depending on the requirement of system. Selection of best algorithm depends on criteria like throughput, waiting time, response time and fairness. ★

Study Tip

Draw Gantt Chart First, then calculate WT, TAT & RT.



Q2. Discuss the time-sharing system in detail. Explain its characteristics, advantages and disadvantages.

Ans.

1. INTRODUCTION :

Time-sharing system allows multiple users to use the system simultaneously by sharing the CPU through small time slices called time quantum. It is an extension of multi-programming system.

2. CHARACTERISTICS :

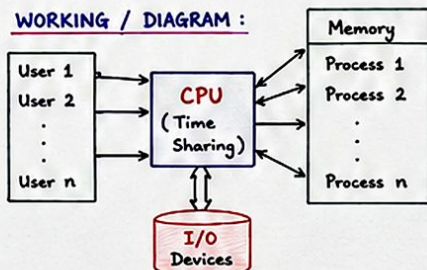
- Many users can access the system at the same time.
- CPU time is divided into small time slices (quantum).
- Each user gets the response quickly.
- It is interactive in nature.
- Preemptive scheduling is used.
- Requires more memory and protection.

Key Point

Time-sharing = CPU time divided into small slices.



3. WORKING / DIAGRAM :



4. ADVANTAGES :

- ✓ Fast response to users.
- ✓ Fairness among users.
- ✓ Increased CPU utilization.
- ✓ Users feel they have dedicated machine.
- ✓ Easy to add new users.

5. DISADVANTAGES :

- ✗ Higher memory requirement.
- ✗ More CPU overhead for context switching.
- ✗ Security and protection problems.
- ✗ Complex to design.

Remember

Small Time Slice
→ Better Response for Users.

6. CONCLUSION : Time-sharing system improves user interaction and CPU utilization by dividing CPU time into small time slices. It is widely used in multi-user environments like universities, banks, and online systems. ★

Quick Revision

UNIT-1 MAP



Very Important Topics ★

- CPU Scheduling (All Algorithms)
- Scheduling Criteria
- Time-sharing System
- Multiprogramming
- I/O Devices, Buffering
- System Calls

Exam Strategy

- ✓ Draw diagrams wherever possible.
- ✓ Write points in proper order.
- ✓ Use headings & underlines.
- ✓ Give examples.
- ✓ Write conclusion at the end.

All The Best



UNIT 1 - OPERATING SYSTEM INTRODUCTION, CPU SCHEDULING & I/O DEVICES

PYQ ALERT

- 12 Marks
- Repeated
- High Priority

★ UNIT 1 TOPICS (SYLLABUS MATCH)

Introduction to OS, OS Services, Multiprogramming, Time-sharing System, Storage Structures, System Calls, Multiprocessor System, Basic CPU Scheduling Concepts, Scheduling Criteria, Scheduling Algorithms, Algorithm Evaluation, Multiple Processor Scheduling, Real-time Scheduling, I/O Devices Organization, I/O Buffering.

MEMORY TRICK

"INTOS SMS":

- Introduction
- Scheduling
- I/O

(Flow of Unit-1) 😊

① Q. Explain CPU Scheduling, its criteria and various scheduling algorithms.

PYQ ALERT: ★ Repeated

Year: 2016, 2017, 2019, 2021, 2023

Marks: 12

ANS:

Introduction:

CPU Scheduling is the process of selecting a waiting process from the ready queue and allocating the CPU to it. It is important to keep CPU busy and improve system performance.

Need:

- Fair allocation of CPU
- High CPU utilization
- Good response time
- Avoid starvation

Scheduling Criteria:

Criteria	Goal
CPU Utilization	Maximize
Throughput	Maximize
Turnaround Time	Minimize
Waiting Time	Minimize
Response Time	Minimize

No single algorithm is best for all criteria.
It is a trade-off. ★

Scheduling Algorithms:

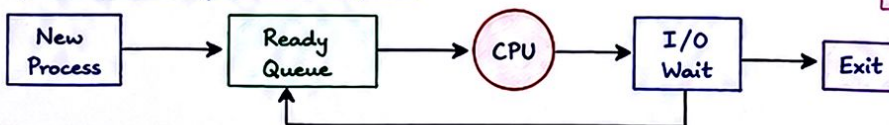
1. First Come First Serve (FCFS)
2. Shortest Job First (SJF)
3. Shortest Remaining Time First (SRTF)
4. Priority Scheduling
5. Round Robin (RR)
6. Multilevel Queue Scheduling
7. Multilevel Feedback Queue Scheduling
8. Real-time Scheduling

Algorithm Evaluation:

Algorithms are compared using Scheduling Criteria.

Example: Avg Waiting Time, Throughput, Turnaround Time.

CPU Scheduling Basic Diagram:



Scheduling is done when:

- Process creates
- Process terminates
- Process goes for I/O
- Time slice expires

② Q. Explain I/O Devices Organization and I/O Buffering.

PYQ ALERT: ★ Repeated

Year: 2015, 2018, 2020, 2022, 2024

Marks: 12

ANS:

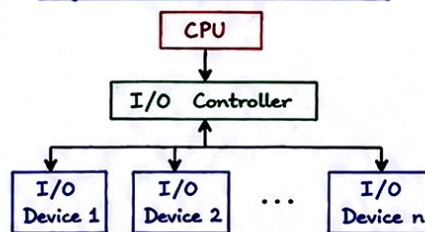
I/O Devices Organization:

I/O devices varies widely in their function, data rate, data form and control methods. The OS must provide uniform interface for these devices.

Types of I/O Devices:

1. Block Devices (e.g., Disk, SSD)
2. Character Devices (e.g., Keyboard)
3. Network Devices
4. Communication Devices
5. Storage Devices
6. Special Purpose Devices

Organization of I/O Devices:



I/O Controller acts as an interface between CPU/Memory and I/O Devices. It has control logic, data buffers and device status registers. ★

I/O Buffering:

Buffering is used to match the difference in speed between CPU and I/O devices. It improves efficiency by reducing the number of I/O operations.

Types of Buffering:

1. Single Buffering
2. Double Buffering
3. Circular Buffering
4. Spooling (Simultaneous Peripheral Operations On-line)

★ IMPORTANT POINTS

- CPU Scheduling is the heart of OS.
- Good scheduling = High performance.
- I/O system is the bridge between CPU and real world.

EXAM TIPS

- Draw diagrams clearly.
- Write criteria in table.
- Give examples wherever possible.
- Write conclusion at end.

QUICK RECALL

FCFS → Simple but not optimal
SJF → Min Waiting Time
RR → Fair to all
Priority → Higher priority first
Multilevel → Different queues
Real-time → Deadline based

CONCLUSION

CPU Scheduling and I/O Organization are essential for efficient functioning of Operating System. They directly affect performance and user satisfaction. ★

Remember: "Better Scheduling → Better Performance → Better System" 😊

UNIT 2 - PROCESSES MANAGEMENT, SYNCHRONIZATION & DEADLOCK

PYQ ALERT

- 12 Marks
- Repeated
- High Priority

★ UNIT 2 TOPICS (SYLLABUS MATCH)

Process Concept, Operations on Processes, Threads, IPC (Inter Process Communication), Precedence Graphs, Semaphores, Classical Synchronization Problems, Deadlock, Deadlock Handling.

MEMORY TRICK

"P O T I P - S D D"

P → Process Concept
O → Operations
T → Threads
I → IPC
P → Precedence Graphs
S → Semaphores
D → Deadlock
D → Deadlock Handling

① Q. Explain Deadlock, its necessary conditions and Deadlock Handling.

PYQ ALERT: ★ Repeated Year: 2016, 2018, 2021, 2023 Marks: 12

ANS:

Introduction:

Deadlock is a situation where a set of processes are blocked forever, each waiting for a resource held by another process in the set.

Example:

P1 holds R1 and waits for R2
P2 holds R2 and waits for R1
then both are in deadlock.

Necessary Conditions (Coffman Conditions):

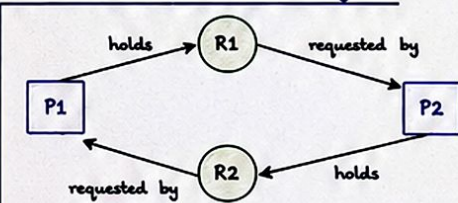
1. **Mutual Exclusion** - At least one resource must be non-shareable.
2. **Hold and Wait** - A process holding at least one resource is waiting for additional resources.
3. **No Preemption** - Resources cannot be preempted; they can be released only voluntarily.
4. **Circular Wait** - A set of processes $\{P_0, P_1, \dots, P_n\}$ must be waiting for resources held by the next process in circular order.

If any one of these conditions is prevented, Deadlock cannot occur. ★

Deadlock Handling Methods:

1. **Deadlock Prevention**
Ensure that at least one of the necessary conditions never holds.
2. **Deadlock Avoidance**
Allocate resources in such a way that the system never enters unsafe state.
3. **Deadlock Detection**
Allow deadlock to occur, then detect it.
4. **Deadlock Recovery**
Recover the system from deadlock by terminating processes or preempting resources.

Deadlock (Circular Wait) Diagram:



EXAM TIP

Always write all 4 conditions for 11-12 marks answers.

Conclusion:

Deadlock is a serious problem in OS. It should be handled carefully using prevention, avoidance, detection and recovery to ensure system efficiency.

② Q. Explain Banker's Algorithm for Deadlock Avoidance with example.

PYQ ALERT: ★ Repeated Year: 2017, 2019, 2022 Marks: 12

ANS:

Introduction:

Banker's Algorithm is a resource allocation and deadlock avoidance algorithm. It ensures that the system never enters an unsafe state by granting resource requests only if the system remains in safe state.

Data Structures:

- Available[m] - No. of available resources of each type.
- Max[n][m] - Maximum demand matrix.
- Allocation[n][m] - Currently allocated resources.
- Need[n][m] = Max - Allocation

Algorithm:

1. Work = Available
2. Finish[i] = false for all processes
3. Find i such that Finish[i] = false and Need[i] ≤ Work
4. If found, Work = Work + Allocation[i]; Finish[i] = true, repeat step 3.
5. If Finish[i] = true for all i, system is in safe state.
6. If no such i found, system is in unsafe state.

Safety Check Flow:



Example:

Let there are 5 processes P₀-P₄ and 3 resource types A, B, C.

	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P ₀	0	1	0	7	5	3			
P ₁	2	0	0	3	2	2			
P ₂	3	0	2	9	0	2	3	3	2
P ₃	2	1	1	2	2	2			
P ₄	0	0	2	4	3	3			

Need = Max - Allocation

	Need		
	A	B	C
P ₀	7	4	3
P ₁	1	2	2
P ₂	6	0	0
P ₃	0	1	1
P ₄	4	3	1

Safe Sequence:

< P₁, P₃, P₄, P₀, P₂ >

Since all processes can finish in this order, the system is in safe state.

IMPORTANT POINTS

- Banker's Algorithm = Deadlock Avoidance.
- It always keeps system in safe state.
- Used in systems with multiple resource types.

MEMORY TRICK

"Never Grant If Unsafe"
• Check Need ≤ Work
• Then Only Grant
• Keep System SAFE

QUICK FORMULA

Need[i][j] = Max[i][j] - Allocation[i][j]
Safe State ⇒ All Finish[i] = true
Unsafe State ⇒ Some Finish[i] = false

EXAM TIP

- Draw table neatly.
- Show all steps clearly.
- Conclude with Safe or Unsafe state.

Remember: "Prevent Deadlock, Avoid Deadlock, Detect Deadlock, Recover Smartly"

UNIT-2

BCA-205 : OPERATING SYSTEM & UNIX

PROCESS MANAGEMENT, SYNCHRONIZATION & DEADLOCK

PYQ ALERT

- 12 Marks
- Repeated
- High Priority ★

UNIT-2 TOPICS (SYLLABUS MATCH)

- ★ Process concept, Operations on processes, Threads, IPC (Inter Process Communication), Precedence graphs, Semaphores, Classical synchronization problems, Deadlock, Deadlock handling.

MEMORY TRICK

PSS =
Process
Synchronize
to
Stop
Deadlock 😊

① Q. Explain Process Concept and Operations on Processes.

PYQ ALERT : ★ Repeated

Year : 2016, 2018, 2021, 2023

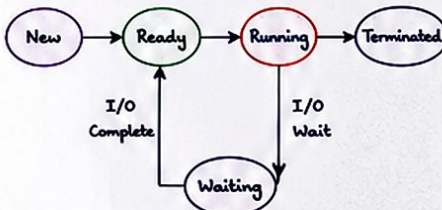
Marks : 12

ANS:

Introduction:

A process is a program in execution. It is the basic unit of CPU utilization. Each process has its own address space, data, code, stack, program counter and resources.

Process State :



Operations on Processes :

1. Create Process
2. Terminate Process
3. Suspend Process
4. Resume Process
5. Block Process
6. Unblock Process
7. Change Process State
8. Inter Process Communication
9. Context Switch

Explanation :

- Create : OS allocates PCB and resources.
- Terminate : OS releases all resources.
- Suspend : Process is moved to waiting state.
- Resume : Suspended process becomes ready.
- Block : Process waits for an event.
- Unblock : Event occurs, process becomes ready.
- Context Switch : CPU moves from one process to another.

IMPORTANT POINTS

- PCB (Process Control Block) contains all information of a process.
- Process changes state based on CPU scheduling and I/O events.
- Multiprogramming increases CPU utilization.

EXAM TIP :

Always draw process state diagram and list operations in order. 💡

② Q. Explain Inter Process Communication (IPC) and its methods.

PYQ ALERT : ★ Repeated

Year : 2017, 2019, 2022, 2024

Marks : 12

ANS:

Introduction:

IPC allows processes to exchange data and information and to coordinate their actions. It is essential in a system when multiple processes are running concurrently.

Methods of IPC :

1. Shared Memory
2. Message Passing
3. Pipes
4. Signals
5. Sockets (in UNIX)

1) Shared Memory :

A region of memory is shared among processes. No data copying. Fast but needs synchronization.



2) Message Passing :

Processes communicate by sending messages through messages.

Types :

- Direct Communication
- Indirect Communication (Mailbox)

3) Pipes :

Unidirectional communication channel. Used in parent-child processes.

Types :

- Ordinary Pipe
- Named Pipe (FIFO)

4) Signals :

Used to notify a process about an event.

Example : SIGINT, SIGTERM, SIGKILL.

5) Sockets :

Used for communication over a network in UNIX systems.

GOOD TO REMEMBER

Shared Memory = Fast
Message Passing = Safe
Pipes = Simple
Signals = Notification
Sockets = Network IPC

③ Q. Explain Deadlock and its Handling Methods.

PYQ ALERT : ★ Repeated

Year : 2015, 2017, 2020, 2023

Marks : 12

ANS:

Introduction:

Deadlock is a situation where a set of processes are blocked because each process is holding a resource and waiting for another resource acquired by some other process in the set.

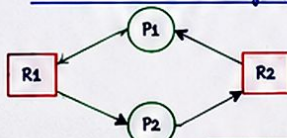
Necessary Conditions (Coffman Conditions):

1. Mutual Exclusion
2. Hold and Wait
3. No Preemption
4. Circular Wait

Deadlock Handling Methods :

1. Deadlock Prevention (Break one of the four conditions)
2. Deadlock Avoidance (Resource Allocation Graph / Banker's Algorithm)
3. Deadlock Detection
4. Deadlock Recovery

Resource Allocation Graph (RAG):



R → Resource
P → Process
Red → Request Edge
Green → Assignment Edge

Banker's Algorithm (Avoidance):

System is in safe state if there exists a sequence in which all processes can get their needed resources and finish.

EXAM TIP

Define deadlock → Write four conditions → Draw RAG → Write handling methods.

QUICK RECALL : IPC = Communication between processes Deadlock = Wait forever for resources ★

- 12 Marks
- Repeated
- High Priority

★ **UNIT 3 TOPICS (SYLLABUS MATCH)**

Basic concepts, Logical vs Physical address space, Contiguous memory allocation, Paging, Segmentation, Demand Segmentation, Page Replacement Algorithms (FIFO, LRU, Optimal).

MEMORY TRICK

"Pages are like book pages."
LRU - Least Recently Used ☺

① **Q. Explain the concept of Paging in detail. Describe address translation in paging.**

PYQ ALERT: ★ Repeated

Year: 2016, 2018, 2020, 2022

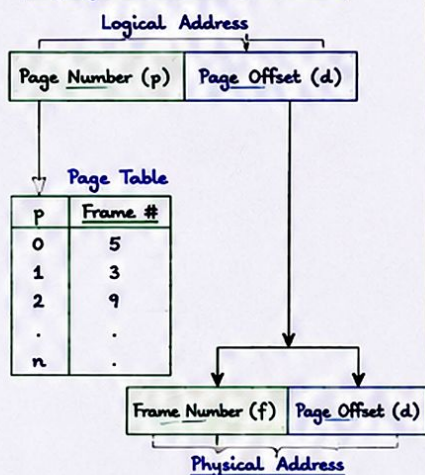
Marks: 12

ANS:Introduction:

Paging is a memory management scheme in which the logical address space of a process is divided into fixed-size blocks called pages and physical memory is divided into fixed-size blocks called frames.

Explanation:

- Pages of a process can be placed in any free frame.
- Size of page = size of frame.
- Eliminates external fragmentation.
- May cause internal fragmentation in last page.

Paging Address Translation:Example:

Page size = 1 KB (2^{10} bytes)
 Logical address = 3200
 Page number (p) = $3200 / 1024 = 3$
 Page offset (d) = $3200 \% 1024 = 128$
 If page table[3] = 7 → Frame = 7
 Physical address = $(7 \times 1024) + 128 = 7296$

Address Translation Steps:

1. CPU generates logical address (p, d).
2. p is used as index into page table.
3. Get frame number (f).
4. Physical address = (f, d)
5. If entry not present → Page Fault.

Important Points:

- Page table keeps track of all pages.
- Large page table → More memory.
- TLB (Translation Lookaside Buffer) is used for fast lookup.
- Paging supports virtual memory.

Conclusion:

Paging is an efficient memory management technique that allows non-contiguous allocation and supports virtual memory. ★

② **Q. Explain Page Replacement Algorithms with example. Discuss FIFO, LRU and Optimal algorithms.**

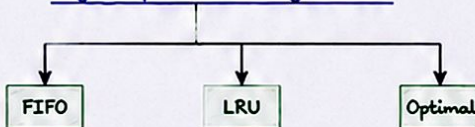
PYQ ALERT: ★ Repeated

Year: 2015, 2017, 2019, 2021, 2023

Marks: 12

ANS:Introduction:

When a page referenced by a process is not in memory, a page fault occurs. If no free frame is available, we must replace one of the pages in memory using a page replacement algorithm.

Page Replacement Algorithms:Important Points:

- These algorithms are used when page fault occurs and no free frame is available.
- Goal → Minimize page faults.

(A) FIFO (First In First Out):

- The page that has been in memory the longest is replaced first.
- Simple to implement.
- May suffer Belady's Anomaly.

Example:

Reference string: 7 0 1 2 0 3 0 4 2 3 0 3 2
 No. of frames = 3

F1	7	7	7	2	2	2	4	4	0	0	2	2
F2		2	2	2	2	2	2	2	2	2	2	2
F3		2	2	2	2	2	2	2	2	2	2	2
PF	F	F	F	F	F	F	F	F	F	F	F	F

Page Faults = 9

(B) LRU (Least Recently Used):

- The page that has not been used for the longest time is replaced.
- Better than FIFO.
- No Belady's Anomaly.

Example:

Reference string: 7 0 1 2 0 3 0 4 2 3 0 3 2
 No. of frames = 3

F1	7	7	7	2	2	0	0	4	4	0	2	2
F2		2	2	2	2	2	2	2	2	2	2	2
F3		2	2	2	2	2	2	2	2	2	2	2
PF	F	F	F	F	F	F	F	F	F	F	F	F

Page Faults = 7

(C) Optimal:

- The page that will not be used for the longest period in future is replaced.
- Gives minimum page faults.
- Not possible to implement in real systems (theoretical).

Example:

Reference string: 7 0 1 2 0 3 0 4 2 3 0 3 2
 No. of frames = 3

F1	7	7	7	2	2	0	0	4	4	0	2	2
F2		2	2	2	2	2	2	2	2	2	2	2
F3		2	2	2	2	2	2	2	2	2	2	2
PF	F	F	F	F	F	F	F	F	F	F	F	F

Page Faults = 6

Comparison Table:

Algorithm	Advantage	Disadvantage
FIFO	Simple	More page faults, Belady's anomaly
LRU	Better performance	Complex, needs time tracking
Optimal	Minimum page faults	Not implementable

Conclusion:

Page replacement algorithms are essential for efficient use of memory. LRU is more practical and gives better results than FIFO. ★

Memory Trick:

FIFO → First In First Out (Queue)

LRU → Least Recently Used (Time)

OPT → Best Possible (Future) 💡

- 12 Marks
- Repeated
- High Priority

UNIT 3 TOPICS (SYLLABUS MATCH)

Background, Logical vs Physical Address Space, Memory Management Unit (MMU), Relocation, Swapping, Contiguous Memory Allocation, Paging, Address Translation Scheme, Page Table, Structure of the Page Table, Inverted Page Table, Page Replacement Algorithms (FIFO, Optimal, LRU), Thrashing, Demand Segmentation.

MEMORY TRICK

"BACK LOGIC MMU"

REL-SWAP-CONTIG
PAGE-TABLE-INVERT
REPLACE-THRASH-DEMAND

① Q. Explain Paging and Address Translation Scheme.

PYQ ALERT : ★ Repeated Year: 2015, 2017, 2019, 2021, 2023 Marks: 12

ANS:

Introduction:

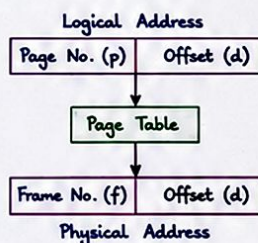
Paging is a memory management scheme in which the logical address space of a process is divided into fixed size blocks called pages and the physical memory is divided into fixed size blocks called frames.

Need of Paging:

- Eliminates external fragmentation.
- Allows non-contiguous allocation.
- Simplifies memory management.

Address Translation Scheme:

CPU generates a logical address (Page number + Offset). MMU uses page table to map page number to frame number. Then physical address is formed by combining frame number with offset.



Page Table Structure:

Page table contains frame number for each page. It is maintained by OS for each process.

Page No.	Frame No.
0	5
1	2
2	9
⋮	⋮
n	f

Size of page table = (Number of pages) × (Size of entry)

Example:

If page size = 4KB, logical address = (p=5, d=3000) and page table[5] = 8 then physical address = (f=8, d=3000)

Conclusion: Paging with address translation allows efficient use of memory and removes external fragmentation.

② Q. Explain Page Replacement Algorithms. Write FIFO, Optimal and LRU with example.

PYQ ALERT : ★ Repeated Year: 2014, 2016, 2018, 2020, 2022 Marks: 12

ANS:

(i) FIFO (First-In First-Out):

The page that has been in memory the longest is replaced first. It is simple but may give more page faults.

Example: Frames = 3

Reference String: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2

Ref	7	0	1	2	0	3	0	4	2	3	0	3	2
F1	7	7	7	2	2	2	4	4	4	0	0	0	0
F2	-	0	0	3	3	3	2	2	3	3	3	3	3
F3	-	-	1	1	1	1	0	2	2	0	0	2	2

Page Faults = 9

(ii) OPTIMAL (MIN):

The page which will not be used for the longest period in future is replaced. It gives minimum page faults but cannot be implemented in OS.

Example: Frames = 3

Reference String: Same as above

Ref	7	0	1	2	0	3	0	4	2	3	0	3	2
F1	7	7	7	7	0	0	4	4	4	4	4	4	2
F2	-	0	0	0	3	3	3	3	3	3	3	3	0
F3	-	-	1	1	1	1	1	2	2	2	2	3	3

Page Faults = 7

(iii) LRU (Least Recently Used):

The page that has not been used for the longest time in the past is replaced. It gives better result than FIFO.

Example: Frames = 3

Reference String: Same as above

Ref	7	0	1	2	0	3	0	4	2	3	0	3	2
F1	7	7	7	2	2	2	4	4	4	0	0	0	0
F2	-	0	0	0	3	3	3	3	3	3	3	3	3
F3	-	-	1	1	1	1	2	2	2	2	2	2	2

Page Faults = 8

Note: OPTIMAL ≤ LRU ≤ FIFO in terms of Page Faults.

IMPORTANT POINTS

- FIFO is easy but not efficient.
- OPTIMAL gives minimum faults.
- LRU approximates optimal.
- More page faults → More I/O → Less performance.

③ Q. What is Thrashing? Explain it. How can it be controlled?

PYQ ALERT : ★ Repeated Year: 2016, 2018, 2020, 2022 Marks: 12

ANS:

Thrashing:

Thrashing is a condition in which system spends most of its time in paging (swapping pages in and out) instead of executing processes due to insufficient frames. It leads to very low CPU utilization and poor system performance.

EXAM TIP:

Write definition → reasons → effects → control techniques → conclusion. Diagram helps.

Causes of Thrashing:

- High degree of multiprogramming.
- Insufficient number of frames.
- Poor locality of reference.

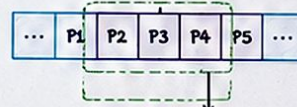
Effects:

- High paging activity.
- Low CPU utilization.
- Degraded performance.
- Long response time.

How to Control Thrashing:

1. Working Set Model - Allocate frames based on working set of process.
2. Page Fault Frequency (PFF) - If page fault rate is high, allocate more frames.
3. Reduce Degree of Multiprogramming.
4. Increase Physical Memory.
5. Use Locality of Reference.

Working Set Window



Frames allocated based on working set

Conclusion: Thrashing occurs when processes do not have enough pages in memory. It can be controlled by proper memory allocation and by using models like Working Set and PFF.

UNIT-4 FILE SYSTEM & DISK MANAGEMENT

(FILE CONCEPTS, DIRECTORY STRUCTURE, FILE OPERATIONS, DISK MANAGEMENT, VIRTUAL FILE SYSTEM)

PYQ ALERT

- 12 Marks
- Repeated
- High Priority

★ UNIT-4 TOPICS (SYLLABUS MATCH)

File concept, File attributes, File operations, File types, Access methods, Directory structure, Disk structure, Free space management, Disk scheduling, Mounting, Virtual File System (VFS)

MEMORY TRICK

"Files are stored in DISK, accessed by OS, managed by File System. ☺"

① Q. Explain File System in detail. Describe file operations and directory structure.

PYQ ALERT: ★ Repeated Year: 2015, 2018, 2020, 2022 Marks: 12

ANS:

Introduction:

A file system is the part of an operating system that manages files and directories on secondary storage. It provides efficient storage, retrieval and protection of data.

File Attributes:

1. Name
2. Identifier
3. Type
4. Location
5. Size
6. Protection
7. Time, date, user ID

File Operations:

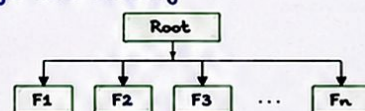
1. Create - create a new file
2. Open - open an existing file
3. Read - read data from file
4. Write - write data into file
5. Append - add data at end
6. Close - close the file
7. Delete - remove file from system
8. Truncate - remove all data but keep file attributes

IMPORTANT POINTS

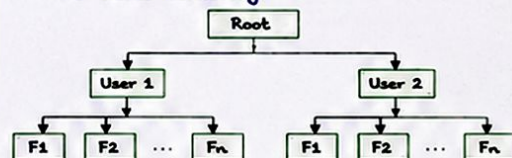
- File is a collection of related information.
- Directory is a special file that contains information about other files.
- File system provides naming, organization, security and space management.

Directory Structure:

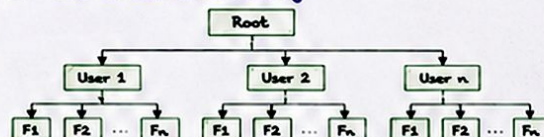
1. Single Level Directory



2. Two Level Directory



3. Tree Structured Directory



② Q. Explain Disk Management in detail. Describe disk scheduling algorithms.

PYQ ALERT: ★ Repeated Year: 2016, 2019, 2021, 2023 Marks: 12

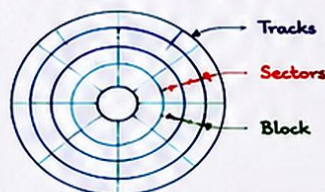
ANS:

Disk Management:

Disk management includes allocation of disk blocks, free space management, disk scheduling and disk structure to improve performance.

1. Disk Structure:

- Disk consists of platters.
- Each platter has two surfaces.
- Each surface is divided into tracks.
- Each track is divided into sectors.
- Block = one or more sectors.



TERMS TO REMEMBER

Track → Circular path on disk surface
Sector → Smallest unit of data transfer
Block → One or more sectors
Cylinder → Set of same tracks on all surfaces

2. Free Space Management:

Keeps track of free and used disk blocks.

Methods:

- (a) Bit Vector
- (b) Linked List
- (c) Grouping
- (d) Counting

3. Disk Scheduling Algorithms:

Disk scheduling decides the order in which disk service requests are satisfied.
Goal → Minimize seek time.

(a) FCFS (First Come First Serve)

- Simple, fair
- Poor performance

(b) SSTF (Shortest Seek Time First)

- Choose request with minimum seek time
- Good performance

(c) SCAN (Elevator Algorithm)

- Move in one direction, serve requests then reverse

(d) C-SCAN (Circular SCAN)

- Move in one direction only
- After reaching end, jump to beginning

(e) LOOK

- Like SCAN but does not go to end if no request

Example (SCAN Algorithm)

Request Queue: 98, 183, 41, 122, 14, 124, 65, 67
Initial Head Position = 53, Direction = High



Seek Sequence: 53 → 65 → 67 → 98 → 122 → 124 → 183 → 199 → 41 → 14

Total Seek = |65-53| + |67-65| + |98-67| + |122-98| + |124-122| + |183-124| + |199-183| + |199-41| + |41-14| = 640

IMPORTANT POINTS

- Disk I/O is slower than main memory.
- Good disk scheduling reduces seek time.
- Free space management increases efficiency.
- Proper directory and file organization improves system performance.

EXAM TIP

For any disk scheduling problem → Draw the number line first and then calculate total seek time step by step.

QUICK RECALL: File → Stored on Disk → Managed by File System → Organized in Directories → Accessed by OS → Efficient I/O with Disk Management

PROMPT FOR NEXT UNIT (COPY & PASTE)

Use the same Prompt and just change the text as below for next unit.

For UNIT-1
(Already used)

For UNIT-2
Change "UNIT 1 ONLY."
to "UNIT 2 ONLY."

For UNIT-3
Change "UNIT 1 ONLY."
to "UNIT 3 ONLY."

For UNIT-4
(This page)

For UNIT-5
Change "UNIT 1 ONLY."
to "UNIT 5 ONLY."

For UNIT-6
Change "UNIT 1 ONLY."
to "UNIT 6 ONLY."

Always keep: 12-MARK PYQs ONLY + Repeated/High Frequency + UNIT WISE ONLY

UNIT-4 : FILE MANAGEMENT SYSTEM

(File Concept , Directory Structure , File System , File Access Methods , File Operations , File Protection)

PYQ ALERT ★

- 12 Marks
- Repeated
- High Priority 😊

★ UNIT-4 TOPICS (SYLLABUS MATCH)

File concept , File attributes , File types , File operations , Directory structure (single level, two level, tree, acyclic graph, general graph) , File system Mounting , File access methods , File protection.

MEMORY TRICK

"File System" means
Files + Directory + Protection 💡

① Q. Explain File System in detail.

PYQ ALERT : ★ Repeated

Year : 2015, 2016, 2018, 2021

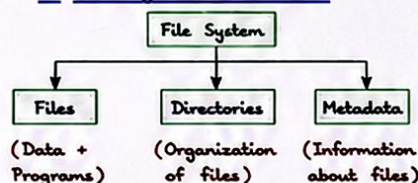
Marks : 12

ANS:

1. Introduction:

A file system is the logical structure and method used by an operating system to store, manage, organize and retrieve files on secondary storage.

2. File System Structure:



5. File Operations:

- Create : Create a new file.
- Open : Open an existing file.
- Read : Read data from file.
- Write : Write data into file.
- Close : Close an opened file.
- Delete : Remove a file.

3. Components of File System:

- File : Collection of related information.
- Directory : Special file that contains information about other files.
- File Attributes : Name, type, size, location, protection, time, owner etc.
- File Operations : Create, delete, open, close, read, write, rename.
- File Access Methods : Sequential, Direct, Indexed.
- File Protection : Prevent unauthorized access and ensure data security.

4. Directory Structures:

- Single Level Directory
- Two Level Directory
- Tree Structured Directory
- Acyclic Graph Directory
- General Graph Directory

IMPORTANT POINTS:

- File is a collection of related data.
- Directory helps in organizing files.
- File attributes define file properties.
- Protection ensures data security.
- Access methods improve efficiency.

7. File Protection:

- Access control using passwords.
- Read, Write, Execute permissions.
- Owner, Group, Others protection.
- Prevent unauthorized access.

Conclusion:

File system provides efficient and secure management of data on secondary storage. ★

② Q. Explain Directory Structures with neat diagrams.

PYQ ALERT : ★ Repeated

Year : 2017, 2019, 2022

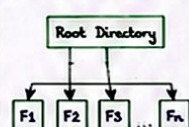
Marks : 12

ANS:

Introduction:

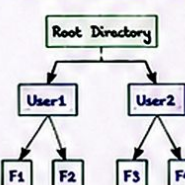
Directory is a special file that contains information about other files. Different types of directory structures are used to organize files.

1. Single Level Directory



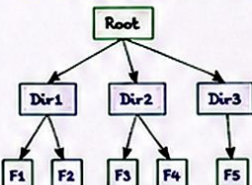
- All files are in one directory.
- Simple but not suitable for large systems.

2. Two Level Directory



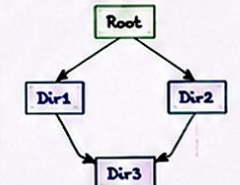
- Each user has own directory.
- Same file name can exist for different users.

3. Tree Structured Directory



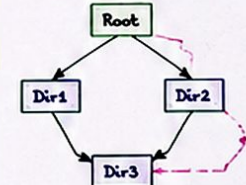
- Hierarchical structure.
- Most widely used.

4. Acyclic Graph Directory



- Allows sharing of files and sub-directories.
- No cycles allowed.

5. General Graph Directory



- Allows sharing with cycles.
- More flexible but complex to manage.

EXAM TIP

Draw neat diagrams and write key points for each structure. It helps in scoring full marks.

Comparison Table:

Directory Type	Structure	Sharing	Cycles Allowed	Complexity	Use
Single Level	All files in one directory	No	No	Low	Small systems
Two Level	User separate directories	No	No	Low	Multi-user systems
Tree	Hierarchical structure	No	No	Medium	General purpose
Acyclic Graph	Shared files & directories	Yes	No	High	Network systems
General Graph	Shared with cycles	Yes	Yes	Very High	Special systems

Memory Trick

S → Single (Simple)
T → Two level (Users)
T → Tree (Hierarchical)
A → Acyclic (No Cycles)
G → General (With Cycles) 😊

★ QUICK RECALL : ★ File = Data Collection, → Directory = Organizer, → File System = Management + Protection 😊

NEXT UNIT
PROMPT (COPY
& PASTE) →

Create a COMPLETE A4 PORTRAIT HANDWRITTEN MICRO-CHEATSHEET for UNIT 5 only, using ONLY my uploaded official syllabus + ALL uploaded PYQs + notes.
(Same rules: 12-MARK PYQs ONLY, Repeated/High Priority, verified years only, neat diagrams, exam-copy answers.)

UNIT-5 : UNIX SYSTEM PROGRAMMING

(SHELL, SHELL SCRIPTING, vi EDITOR, UNIX COMMANDS, TEXT PROCESSING UTILITIES)

PYQ ALERT ★

- 12 Marks
- Repeated.
- High Priority

★ UNIT-5 TOPICS (SYLLABUS MATCH)

Shell concept, Types of Shells, Shell scripting, vi editor, Basic & advanced UNIX commands, Text processing utilities (grep, sed, awk, cut, paste, sort, wc, uniq, tr)

MEMORY TRICK

"Shell is the door,
Script is the key,
Commands are tools,
Utilities make work easy,"

① Q. Explain the UNIX Shell and write a shell script to find the largest of n numbers. Also explain vi editor commands.

PYQ ALERT : ★ Repeated

Year : 2016, 2018, 2020, 2022

Marks : 12

ANS:

(A) UNIX Shell :

A shell is a command interpreter that acts as an interface between the user and the UNIX kernel. It reads commands from the user, interprets them and executes the corresponding programs.

Types of Shells :

- Bourne Shell (sh)
 - C Shell (csh)
 - Korn Shell (ksh)
 - Bourne Again Shell (bash)
- (bash is most commonly used)

(B) Shell Script to find Largest of n numbers :

```
#!/bin/bash
echo -n "Enter how many numbers : "
read n
echo -n "Enter $n numbers : "
read a
max=$a
i=1
while [ $i -lt $n ]
do
    read a
    if [ $a -gt $max ]
    then
        max=$a
    fi
    i=`expr $i + 1`
done
echo "Largest number is : $max"
```

Example :

Input : 5 numbers → 12 45 8 67 23
Output : Largest number is : 67

Explanation :

- Read first number in max.
- Read remaining numbers one by one.
- Compare with max.
- Update max if larger.
- Finally print max.

(C) vi Editor (Important Commands) :

Command	Function
vi filename	Open file in vi
i	Insert mode
Esc	Exit insert mode
:w	Save file
:wq	Save and quit
:q!	Quit without saving
:dd	Delete current line
:set nu	Show line numbers
:/text	Search text
:n	Find next
:x	Save and exit

IMPORTANT POINTS :

- Shell is command interpreter.
- Shell script is saved with .sh extension.
- vi is most powerful text editor in UNIX.
- Practice commands daily.

② Q. Explain UNIX commands and text processing utilities with examples.

PYQ ALERT : ★ Repeated

Year : 2015, 2017, 2019, 2021

Marks : 12

ANS:

(A) Basic UNIX Commands :

Command	Description	Example
ls	List files	ls -l
cd	Change directory	cd /home
pwd	Print working directory	pwd
mkdir	Make directory	mkdir test
rmdir	Remove directory	rmdir test
cp	Copy files	cp a.txt b.txt
mv	Move or rename	mv a.txt b.txt
rm	Remove file	rm a.txt
cat	Display file content	cat file.txt

EXAM TIP

- Write command with correct syntax.
- Give example in exam.
- Diagrams/Flowcharts increase marks.
- Practice in terminal daily.

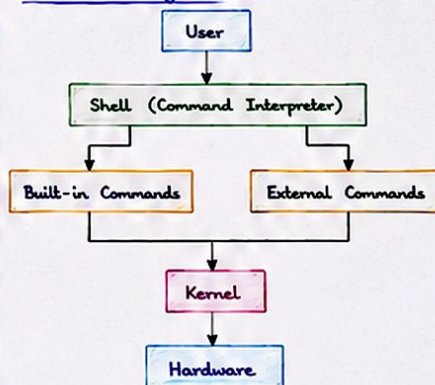
Shell Script Execution :

- Save file : script.sh
- Give permission : chmod +x script.sh
- Execute : ./script.sh

(B) Text Processing Utilities :

1. **grep** → Search pattern in file
Example : grep "error" log.txt
(Searches 'error' in log.txt)
2. **sed** → Stream editor (search & replace)
Example : sed 's/old/new/g' file.txt
(Replace old with new)
3. **awk** → Pattern scanning & processing
Example : awk '{print \$1}' data.txt
(Print first column)
4. **cut** → Extract sections from each line
Example : cut -d: -f1 /etc/passwd
(Cut first field using ':')
5. **sort** → Sort lines of a file
Example : sort names.txt
(Sort files alphabetically)
6. **wc** → Count lines, words, characters
Example : wc -l file.txt
(Count lines)
7. **uniq** → Remove duplicate lines
Example : sort file.txt | uniq
(Remove duplicates)
8. **tr** → Translate or delete characters
Example : tr 'a-z' 'A-Z' < file.txt
(Convert lowercase to uppercase)

Shell Working (Flow) :



IMPORTANT NOTES :

- Shell reads command → interprets → executes program.
- Text utilities work with standard input and output (I/O redirection possible).
- These tools save time and effort.
- These topics are very important for 12-mark questions.

QUICK RECALL (UNIT-5 IN ONE LINE) :

Shell → Command Interpreter | Shell Script → Automate work | vi → Text Editor
Commands → Manage files | Text Utilities → Process text efficiently

PROMPT FOR NEXT UNIT (COPY & PASTE)



Always keep : 12-MARK PYQs ONLY + Repeated/High Frequency + UNIT WISE ONLY

★ UNIT-5 TOPICS (SYLLABUS MATCH)

Shell - Types, Shell Programming Basics, Shell Scripts, Variables, Operators, Control Statements, UNIX Commands (File, Directory, Process, Filters), vi Editor, UNIX System Administration (Users, Groups, Permissions, Backup)

MEMORY TRICK

"SHELL" helps you
"SH" (Start)
"E" (Execute)
"LL" (Logical work)

① Q. Explain UNIX Shell Programming with variables, operators and control structures with examples.

PYQ ALERT : ★ Repeated Year : 2015, 2017, 2019, 2021, 2023 Marks : 12

ANS:

Introduction:

Shell is the command interpreter that acts as an interface between user and kernel. Shell programming means writing a program (script) to perform tasks automatically using shell commands.

1. Shell Variables

- Used to store data.
- No space before & after =
- Types - (i) User defined (ii) Special variables

Example:

```
NAME="AKU"
echo $NAME
```

Example Shell Script

```
#!/bin/sh
# To find largest of two numbers
read A
read B
if [ $A -gt $B ]
then
    echo "A is greater"
else
    echo "B is greater"
fi
```

2. Operators

- Arithmetic : + - * / %
- Relational : -eq -ne -gt -lt -ge -le
- Logical : -a -o !
- String : = != -z -n

Example:

```
[ $A -gt $B ]
```

3. Control Structures

- Sequence - Commands executed one by one.
- Decision - if, if-else, case
- Looping - for, while, until

Example (if-else)

```
if [ $A -gt $B ]
then
    echo "A is greater"
else
    echo "B is greater"
fi
```

IMPORTANT POINTS

- .sh file is a shell script file.
- #! specifies the interpreter.
- chmod +x filename.sh to run.
- Shell scripts save time & reduce human error.

② Q. Explain important UNIX commands with syntax and use.

PYQ ALERT : ★ Repeated Year : 2014, 2016, 2018, 2020, 2022 Marks : 12

ANS:

A. FILE & DIRECTORY COMMANDS

Command	Syntax	Use
ls	ls [option] [file/dir]	List files & directories
cd	cd [directory]	Change directory
pwd	pwd	Print working directory
mkdir	mkdir [dir]	Create new directory
rmdir	rmdir [dir]	Remove empty directory
cp	cp src dest	Copy file
mv	mv src dest	Move/rename file
rm	rm [option] file	Remove file
cat	cat filename	Display file content

B. FILE PERMISSION COMMANDS

Command	Syntax	Use
chmod	chmod mode file	Change file permissions
chown	chown user file	Change file owner
chgrp	chgrp group file	Change group owner

C. PROCESS RELATED COMMANDS

Command	Syntax	Use
ps	ps aux	Show running processes
top	top	Dynamic process view
kill	kill PID	Terminate process
fg	fg	Bring job to foreground
bg	bg	Run job to background

③ Q. Explain vi editor and UNIX system administration.

PYQ ALERT : ★ Repeated Year : 2015, 2017, 2019, 2021, 2023 Marks : 12

ANS:

A. vi EDITOR

vi is a powerful text editor in UNIX.

It has two modes.

1. Command Mode
(Used for navigation)

2. Insert Mode
(Used for typing text)

Common vi Commands

i → Insert mode
Esc → Back to command mode
:w → Save
:q → Quit
:wq → Save & quit
:q! → Quit without saving
dd → Delete current line
yy → Copy (yank) line
p → Paste line
/text → Search forward
:set nu → Show line numbers

B. UNIX SYSTEM ADMINISTRATION

System administration means managing users, groups, permissions and backups to keep system secure & efficient.

1. USER MANAGEMENT COMMANDS

Command	Syntax	Use
useradd	useradd username	Add new user
passwd	passwd username	Set password
userdel	userdel username	Delete user
who	who	Show logged in users
id	id username	Show user info

2. GROUP MANAGEMENT COMMANDS

Command	Syntax	Use
groupadd	groupadd groupname	Create group
groupdel	groupdel groupname	Delete group
groups	groups username	Show user groups

3. PERMISSIONS (r w x)

Symbol	Meaning
r	Read (4)
w	Write (2)
x	Execute (1)

4. BACKUP COMMAND

tar - Used to create backup of files/directories.

Example:

```
tar -cvf backup.tar /home/aku
# Create backup
tar -xvf backup.tar
# Extract backup
```

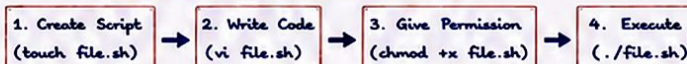
IMPORTANT POINTS

- Super user is 'root'.
- Always use strong password.
- Regular backup prevents data loss.
- Permissions protect system security.

QUICK RECALL

Shell Script → Automate Tasks
UNIX Commands → Manage Files, Processes & System
vi Editor → Edit Files
Admin Commands → Manage Users, Groups & Permissions

SHELL SCRIPT EXECUTION STEPS



EXAM TIP

Write commands with syntax and examples, Draw tables & diagrams where possible.

- 12 Marks
- Repeated
- High Priority

MEMORY TRICK

"U-F-C-V-S-U"

- U - UNIX
- F - File system
- C - Commands
- V - vi editor
- S - Shell
- U - Utilities

★ UNIT-5 TOPICS (SYLLABUS MATCH)

Overview of UNIX, Features, Architecture, File System Hierarchy, File Types, Basic UNIX Commands, vi Editor, Shell Concept, Shell Types, Shell Commands, UNIX Utilities.

① Q. Discuss UNIX Operating System in detail.

PYQ ALERT : ★ Repeated

Year : 2015, 2018, 2020, 2022

Marks : 12

ANS:

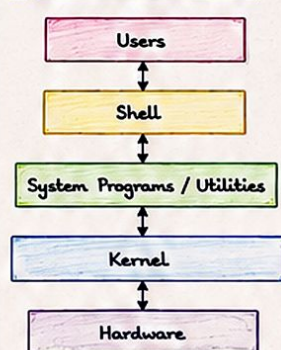
1. Introduction :

UNIX is a powerful, multiuser, multitasking and portable operating system developed at AT&T Bell Labs in 1969 by Ken Thompson, Dennis Ritchie and their team. It is widely used in servers, workstations and embedded systems.

2. Features of UNIX :

- Multiuser - Many users can work simultaneously.
- Multitasking - Many tasks can run at the same time.
- Portability - Can run on different hardware.
- Security - File permissions and user authentication.
- Hierarchical File System - Data organized in tree structure.
- Shell - Powerful command interpreter.

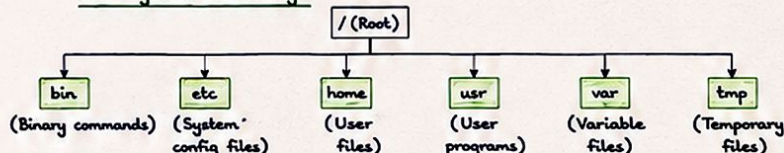
3. Architecture of UNIX :



IMPORTANT POINTS

- UNIX is written mainly in C Language.
- Everything in UNIX is treated as a file.
- Small is beautiful is the UNIX philosophy.
- UNIX supports pipe (|) and redirection (>, >>).

File System Hierarchy



4. File Types in UNIX :

- Regular file
- Directory
- Symbolic link
- Character special file
- Block special file
- Named pipe
- Socket

Example (ls -l Output)

Type	Permission	Links	Owner	Group	Size	Date	Name
	-rw-r--r--	1	zoha	staff	1024	May 10	file.txt
	drwxr-xr-x	2	zoha	staff	4096	May 10	docs
	lrwxrwxrwx	1	zoha	staff	7	May 10	link -> file.txt

Basic UNIX Commands (Most Important)

- ls - list files and directories
- cd - change directory
- pwd - print working directory
- mkdir - make directory
- rmdir - remove directory
- cp - copy files
- mv - move / rename files
- rm - remove files
- cat - display file contents
- more - display file page by page
- less - display file backward / forward
- man - display manual
- who - show logged in users
- date - show system date and time
- cal - show calendar

5. Conclusion :

UNIX is a reliable, secure and efficient OS widely used in real world. Its simplicity, flexibility and power make it unique and still most popular in servers.

QUICK RECALL

UNIX → Multiuser, Multitasking
/ → Root directory
Everything → File
Shell → Command Interpreter

EXAM TIP

Learn commands with options and practice on terminal. Draw file hierarchy neatly in exam for full marks.

② Q. Explain vi editor, Shell and UNIX utilities with examples.

PYQ ALERT : ★ Repeated

Year : 2016, 2019, 2021, 2023

Marks : 12

ANS:

1. vi Editor :

vi is a screen based text editor used to create and edit files in UNIX.

It has three modes.

- Command mode
- Insert mode
- Last line mode

Important vi Commands

- i - insert mode
- Esc - back to command mode
- w - write (save file)
- q - quit
- wq - save and quit
- q! - quit without saving
- dd - delete line
- yy - copy line
- p - paste line
- /text - search forward
- ?text - search backward

2. Shell :

Shell is a command interpreter that acts as an interface between user and UNIX kernel. It reads commands from user and executes them.

Types of Shell

- Bourne Shell (sh)
- C Shell (csh)
- Korn Shell (ksh)
- Bourne Again Shell (bash)
- Z Shell (zsh)

Shell Special Characters

- * - any number of characters
- ? - any single character
- [] - any one character in bracket
- | - pipe
- > - output redirection
- >> - append redirection

3. UNIX Utilities :

Utilities are small programs that perform specialized tasks.

Utility	Function
grep	Search pattern in a file
sort	Sort lines of a file
uniq	Remove duplicate lines
wc	Count lines, words, characters
cut	Cut sections of each line
paste	Merge lines of files
tr	Translate or delete characters
tee	Read from standard input and write to standard output and file
diff	Compare two files
tar	Tape archive (backup files)
gzip	Compress files

IMPORTANT POINTS

- vi is slow but powerful editor.
- Shell saves time and effort.
- Utilities make UNIX more efficient.

EXAMPLE

```

$ ls -l
$ grep "error" file.txt
$ wc file.txt
$ sort data.txt
    
```

Memory Trick

"vi - write, Shell - tell, Utilities - do all well."

- 12 Marks
- Repeated.
- High Priority

★ **UNIT-6 TOPICS (SYLLABUS MATCH)**

User management (useradd , passwd , userdel , groupadd , usermod),
File permissions & ownership (chmod , chown , umask), File system
administration , Backup & recovery , Important UNIX utilities.

MEMORY TRICK

"U P F B U"

U - Users
P - Permissions
F - File system
B - Backup
U - Utilities



① **Q. Explain UNIX user management and related commands in detail.**

PYQ ALERT : ★ Repeated

Year : 2015 , 2018 , 2021 , 2023

Marks : 12

ANS:

Introduction :

UNIX is a multi-user system. User management is used to create, modify, delete users and groups and control their access to the system. It helps in security and proper resource sharing.

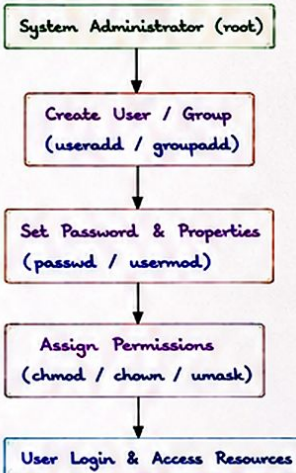
IMPORTANT POINTS

- Every user has UID.
- Root user has UID 0.
- Each user belongs to at least one group.
- /etc/passwd stores user information.
- /etc/shadow stores encrypted passwords.

A) USER MANAGEMENT COMMANDS :

Command	Function
useradd	Create a new user.
passwd	Set or change user password.
userdel	Delete a user.
usermod	Modify user information.
id	Display user identity.
who	Show who is logged in.
whoami	Show current user.
groupadd	Create a new group.
groupdel	Delete a group.
groups	Show groups of a user.

B) Working (Flow)



Example :

1. **useradd zoha**
→ Create user 'zoha'
2. **passwd zoha**
→ Set password for zoha
3. **usermod -aG students zoha**
→ Add zoha to group 'students'
4. **id zoha**
→ Show UID, GID info
5. **userdel zoha**
→ Delete user zoha
6. **groupadd developers**
→ Create new group
7. **groups zoha**
→ Show groups of zoha

Files Related to Users :

- /etc/passwd → User account information
- /etc/shadow → Encrypted passwords
- /etc/group → Group information
- /etc/gshadow → Secure group information

EXAM TIP

- Always write command with syntax.
- Give at least 5-6 examples.
- Draw flow diagram if asked.
- Write files name related to users.

② **Q. Explain file permission and ownership in UNIX. Also explain backup and recovery methods.**

PYQ ALERT : ★ Repeated

Year : 2016 , 2019 , 2022 , 2024

Marks : 12

A) FILE PERMISSIONS :

UNIX uses rwx permission for files and directories.

Permission	Symbol	Value
Read	r	4
Write	w	2
Execute	x	1
No access	-	0

Permission Format (chmod)

rwx	rwx	rwx
User (Owner)	Group	Others

Example : **chmod 754 file.txt**
→ Owner (rwx), Group (r-x), Others (r--)

B) FILE OWNERSHIP :

Every file has an owner and group.

- **chown** → Change owner.
- **chgrp** → Change group.

Example :

chown zoha file.txt
chgrp students file.txt

IMPORTANT POINTS

- Root user can change owner.
- Use numeric mode or symbolic mode in chmod.
- Default permission depends on umask.

C) BACKUP & RECOVERY :

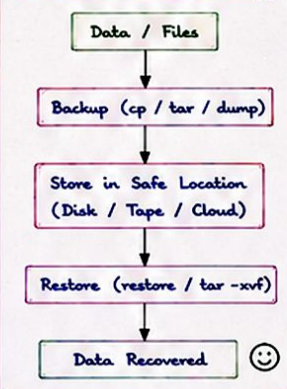
Backup is a copy of data to protect from loss.

- **cp** → Create backup
- **tar** → Backup multiple files
- **dump** → Backup file system
- **restore** → Recover files

Example :

1. **cp -r /home/zoha /backup/**
2. **tar -cvf backup.tar /home/zoha**
3. **tar -xvf backup.tar**
4. **dump -0a -f backup.dump /home**
5. **restore -rf backup.dump**

FLOW (Backup & Recovery)



D) IMPORTANT UNIX UTILITIES :

Command	Use	Example
du	Show disk usage	du -sh /home
df	Show file system space	df -h
find	Search files	find / -name file.txt
locate	Find files (fast)	locate file.txt
which	Show path of command	which ls

Command	Use	Example
ps	Show running processes	ps -ef
top	Real time process info	top
grep	Search text in files	grep 'error' log.txt
wc	Count lines, words, chars	wc -L file.txt
man	Show manual page	man ls

EXAM TIP

- Write command with purpose.
- Add example for every command.
- Draw flowchart for backup & recovery.
- Use tables for permissions & utilities.

QUICK RECALL (UNIT-6 IN ONE LINE) :

Users → Permissions → File System → Backup → Utilities → Strong UNIX Administration



PROMPT FOR NEXT UNIT (COPY & PASTE)

For UNIT-1
(Already used)

For UNIT-2
Change "UNIT 5" to
"UNIT 2 ONLY."

For UNIT-3
Change "UNIT 5" to
"UNIT 3 ONLY."

For UNIT-4
Change "UNIT 5" to
"UNIT 4 ONLY."

For UNIT-5
(This Page)

For UNIT-6
Change "UNIT 5" to
"UNIT 6 ONLY."

UNIT 1 : OS INTRODUCTION, CPU SCHEDULING & I/O

1. OS INTRODUCTION

OS is an interface between user and hardware. It manages all resources and provides services.

OS SERVICES

- Process Management
- Memory Management
- File Management
- I/O Management
- Security & Protection
- Resource Allocation
- Error Detection

2. TYPES OF OS

- Batch OS
- Multiprogramming OS
- Time Sharing OS
- Real Time OS
- Distributed OS
- Network OS

2. OS CONCEPTS

- Multiprogramming
- Time-sharing system
- Storage structures
- System calls
- Multiprocessor system

System Call: Interface to request service from OS.

Types: Process control, File management, Device management, Information maintenance, Communication

3. CPU SCHEDULING ALGORITHMS

Algorithm	Type	Priority	Real	Batch
FCFS	Non	No	No	Yes
SJF	Non	No	No	Yes
RR	Non	Yes	Yes	Yes
Priority	Non	Yes	Yes	Yes
Round Robin	Non	Yes	Yes	Yes
Shortest Job	Non	Yes	Yes	Yes
Longest Job	Non	Yes	Yes	Yes

3. CPU SCHEDULING BASICS

- CPU Scheduling decides which process runs next on CPU.
- Scheduler: Short term scheduler (CPU Scheduler)
- Dispatcher: Gives control of CPU to selected process.
- Context Switch: Saving state of current process and loading state of next process.

4. SCHEDULING CRITERIA

- CPU Utilization - Maximise
- Throughput - Maximise
- Turnaround Time - Minimise
- Waiting Time - Minimise
- Response Time - Minimise
- Fairness - Avoid starvation

5. REAL TIME SCHEDULING

Types:

- Hard Real Time - Missing deadline = System failure
- Soft Real Time - Missing deadline reduces performance.

Algorithms:

- Rate Monotonic (RM)
- Earliest Deadline First (EDF)

8. I/O DEVICES ORGANIZATION

I/O devices connected via I/O controllers.

Types:

- Block Devices (Disk, SSD)
- Character Devices (Keyboard)
- Network Devices
- Communication Devices
- Storage Devices
- Special Purpose Devices

9. I/O BUFFERING

Buffering improves I/O speed.

Types:

- Single Buffering
- Double Buffering
- Circular Buffering
- Spooling (Simultaneous Peripheral Operations On-Line)

★ PYQ ALERT (UNIT-1) : CPU Scheduling algorithms asked every year • Algorithm comparison • Real Time Scheduling • I/O Organization • Buffering types

UNIT 2 : PROCESS MANAGEMENT, SYNCHRONIZATION & DEADLOCK

1. PROCESS CONCEPT

Program in execution is Process.

PCB (Process Control Block) contains: PID, State, PC, Registers, Priority, Memory info, I/O info.

Process States:

New → Ready → Running → (I/O) → Waiting → (I/O) → Ready → Running → (Terminated)

2. OPERATIONS ON PROCESSES

- Create Process
- Terminate Process
- Suspend Process
- Resume Process
- Block / Unblock
- Context Switch

3. THREADS

Lightweight process.

Benefits:

- Responsiveness
- Resource sharing
- Economy

Types:

- User Level Threads
- Kernel Level Threads
- Hybrid Threads

4. IPC (INTER PROCESS COMMUNICATION)

Processes communicate and synchronize with each other.

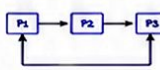
Methods:

- Shared Memory
- Message Passing
 - Direct Communication
 - Indirect Communication

5. PRECEDENCE GRAPHS

Directed graph showing order of resource allocation.

- No cycle → Safe
- Cycle → Deadlock



6. SEMAPHORES

Integer variable used for synchronization.

Types:

- Binary Semaphore (0, 1)
- Counting Semaphore (0..n)

Operations:

- wait(s) / P(s)
- signal(s) / V(s)

8. CLASSICAL SYNC PROBLEMS

- Producer - Consumer
- Readers - Writers
- Dining Philosophers

9. DEADLOCK

Deadlock occurs when set of processes wait forever for resources.

Necessary Conditions:

- Mutual Exclusion
- Hold and Wait
- No Preemption
- Circular Wait

10. DEADLOCK HANDLING

A. Prevention: Violate one of 4 conditions (e.g., No hold & wait)

B. Avoidance: Banker's Algorithm

C. Detection & Recovery: Detect using graph / algorithm. Recover by Kill / Rollback

★ PYQ ALERT (UNIT-2) : Semaphores & Classical problems • IPC methods • Deadlock prevention/avoidance • Banker's Algorithm

UNIT 3 : MEMORY MANAGEMENT & SECURITY

1. MEMORY MANAGEMENT

- Logical & Physical Address Space
- Swapping
- Contiguous Allocation
- Non-Contiguous Allocation
- Paging
- Segmentation
- Paging Segmentation
- Virtual Memory
- Demand Paging
- Page Replacement Algorithms
- Allocation of Frames
- Thrashing
- Demand Segmentation

2. PAGE REPLACEMENT ALGORITHMS

- FIFO (First In First Out)
- LRU (Least Recently Used)
- Optimal (MIN)

Steps:

- Note page reference string
- Frame size
- Apply algorithm
- Count page faults

3. SECURITY

Security protects system from unauthorized access.

Goals:

- Confidentiality
- Integrity
- Availability

4. PROTECTION

Protection controls access to resources.

Mechanisms:

- Access Matrix
- Capability List
- Access Control List

5. INTRUDERS

People who try to break into system.

Types:

- Masquerader
- Misfeasor
- Malicious User

6. COMPUTER VIRUSES

Program that attaches itself to another program.

Types:

- Boot Sector Virus
- File Virus
- Macro Virus
- Worms / Trojan / Spyware

7. TRUSTED SYSTEM

A system in which security mechanisms are correct and effective.

Also called:

- Trusted Computing Base (TCB)
- Secure System

★ PYQ ALERT (UNIT-3) : Paging numerals • Page replacement • Virtual memory • Demand paging • Protection vs Security • Computer Viruses

UNIT 4 : OPEN SOURCE, FILE SYSTEM & LINUX COMMANDS

1. OPEN SOURCE

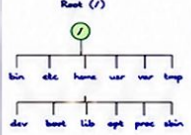
Open Source means software whose source code is freely available to use, modify and distribute.

Advantages:

- Cost Effective
- Flexible
- Secure
- Community Support

2. FILE SYSTEM

- File System Hierarchy
- Logging in



3. LINUX COMMANDS

Command	Use	Command	Use
ls	List files	cp	Copy & convert
cp	Copy files	mv	Move/Rename
mv	Move/Rename	rm	Remove files
rm	Remove files	find	Find files
find	Find files	grep	Search pattern

Directory Commands

Command	Use
mkdir	Create directory
rmdir	Remove directory
cd	Change directory
pwd	Print working directory

Permissions & Links

Command	Use
chmod	Change mode
chgrp	Change group
chown	Change owner

Other Important Topics

- Environment Variables
- PATH setting
- I/O Redirection (>, >>, <)
- Pipes (|)

Ex: ls -l | grep 'txt' > m.txt

UNIT 5 : VI EDITOR, SHELL & LINUX FEATURES

1. VI EDITOR

A screen oriented text editor.

VI MODES:

- Command Mode
- Insert Mode
- Last Line Mode (:)

IMPORTANT COMMANDS

Key	Action
h	Left
j	Down
k	Up
l	Right
i	Insert
o	Open new line
u	Undo
/	Search
q	Quit

2. SHELL

Shell is command interpreter.

Types of Shell:

- Bourne Shell (sh)
- C Shell (csh)
- Korn Shell (ksh)
- Bash Shell (bash)

SHELL FEATURES

- Command Line
- Standard input / output
- Redirection
- Pipes
- Filters
- Special characters (*, ?, [], etc.)
- Searching files / pathnames
- History / shortcuts
- Aliases
- Job control (fg, bg, jobs)
- File substitution
- Shell built-in commands

3. LINUX FEATURES

Drawbacks:

- Not all hardware supported
- Complexity for beginners

Components:

- Kernel
- Shell
- File System
- System Libraries
- Utilities
- Device Drivers

Subsystems:

- Memory Management
- Process Management
- Thread Management
- File Management System

UNIT 6 : LINUX UTILITIES & UNIX SYSTEM ADMINISTRATION

1. LINUX COMMANDS & UTILITIES

- Entering the machine
- User names and groups
- Logging in
- Correcting typing mistakes
- Formed of Linux commands
- Changing password
- Characters with special meanings (*, ?, [], |, >, <)
- Linux documentation (man)
- File system
- Current directory
- Directory contents

2. UNIX SYSTEM ADMINISTRATION

- System Administration
 - User management
 - Group management
 - File permissions
 - Process monitoring
- Installing Linux
 - Choosing installation method
 - Choosing installation class
 - Pre-installation checks
 - Installation steps
- Backing System
 - Boot loader (GRUB)
 - Run levels / Targets
 - Services

3. LINUX UTILITIES

- File Permissions
 - Standard files (*)
 - Output (*)
 - Append (>)
 - Input redirection (<)
 - Output redirection (>)
 - Redirection errors
- File Utilities
 - Standard files (*)
 - Output (*)
 - Append (>)
 - Input redirection (<)
 - Output redirection (>)
 - Redirection errors

4. IMPORTANT FORMULAS

- TAT = CT - AT
- WT = TAT - BT
- RT = First Time CPU - AT
- Avg WT = ΣWT / n
- Avg TAT = ΣTAT / n

5. ONE NIGHT REVISION PLAN

- Unit-1 → OS & CPU Scheduling
- Unit-2 → Processes & Deadlock
- Unit-3 → Memory & Security
- Unit-4 → File System & Linux
- Unit-5 → VI & Shell
- Unit-6 → Admin & Utilities

6. MORNING CHECKLIST

- Revise formulas & shortcuts
- Practice 2-3 numericals
- Revise PYQ questions
- Read short notes
- Stay confident & calm

7. PYQ ALERT (UNIT-6)

- Linux utilities & commands asked regularly
- Permissions, users & file systems important
- Installation, backup & recovery important
- All admin topics asked

8. REMEMBER

Understand → Practice → Revise → Compare with PYQ → Score High.

9. All the Best!

You Can Do It 😊

1. INTRODUCTION TO OPERATING SYSTEM

Q.1 Define Operating System. What are its objectives? (2 Marks) [PYQ ALERT: 2014, 2016, 2022]

Ans. An Operating System (OS) is system software that acts as an interface between user and hardware and manages all resources of the computer.

Objectives :

1. Convenience - make computer easy to use.
2. Efficiency - utilize resources efficiently.
3. Ability to evolve - OS should be flexible to add new functions.

Q.2 List any four functions of Operating System. (2 Marks) [PYQ ALERT: 2017, 2019, 2021]

- Ans.
1. Process Management
 2. Memory Management
 3. File Management
 4. I/O Management
- (Any other valid four)

Q.3 Differentiate between System Software and Application Software. (5 Marks) [PYQ ALERT: 2016]

System Software	Application Software
It manages computer hardware and system resources.	It is used to solve specific user problems.
Examples: OS, Compiler, Assembler, DBMS.	Examples: MS Word, Tally, Photoshop.
Runs in background.	Runs in foreground.
Essential for system operation.	Not essential for system operation.

Q.4 What is the role of OS in a computer system? (5 Marks) [PYQ ALERT: 2018]

Ans. OS acts as a resource manager and control program. It allocates and deallocates resources to users and programs. It controls program execution, handles I/O operations, manages files and memory, ensures security and protection and provides a convenient interface.

2. OPERATING SYSTEM SERVICES

Q.1 Write any five services provided by OS. (2 Marks) [PYQ ALERT: 2016, 2019]

- Ans.
1. Program Execution
 2. I/O Operations
 3. File System Manipulation
 4. Communication
 5. Error Detection
- (Any other)

Q.2 Explain File System Manipulation. (5 Marks) [PYQ ALERT: 2018]

Ans. OS provides mechanism to create, delete, read, write, rename and organize files and directories. It maintains file attributes, access rights and handles storage allocation.

Q.3 What is the need of OS services? (5 Marks) [PYQ ALERT: 2022]

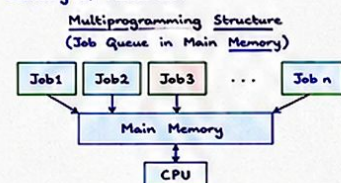
Ans. OS services make the system convenient to use, ensure efficient use of hardware, protect data and resources, provide standard interface for I/O and enable users to share data.

3. MULTIPROGRAMMING

Q.1 Define Multiprogramming and its advantage. (2 Marks) [PYQ ALERT: 2016, 2018, 2021]

Ans. In multiprogramming, more than one program is kept in memory and CPU switches among them to maximize utilization.

Advantage: High CPU utilization, high throughput, economy of resources.



Important Terms

- OS - Interface between user & hardware
- Kernel - Core part of OS
- System Call - Request to OS service
- Process - Program in execution
- CPU Scheduling - Select next process.
- I/O - Input / Output

4. TIME-SHARING SYSTEM

Q.1 Define Time-sharing system. (2 Marks) [PYQ ALERT: 2016, 2017, 2018]

Ans. Time-sharing system is extension of multiprogramming in which CPU time is divided into small time slices (quantum) and each user gets one slice in turn.

Q.2 Advantage of Time-sharing system. (2 Marks) [PYQ ALERT: 2019, 2022]

- Ans.
1. Quick response
 2. Good interactive performance
 3. Fair CPU allocation to users



Each process gets CPU for small time slice.

5. STORAGE STRUCTURES

Q.1 List storage structures used by OS. (2 Marks) [PYQ ALERT: 2018, 2022]

- Ans.
1. PCB (Process Control Block)
 2. Ready Queue
 3. Device Queue
 4. File Control Block (FCB)
 5. Job Queue

Q.2 Explain Process Control Block. (5 Marks) [PYQ ALERT: 2016, 2019]

Ans. PCB is a data structure that contains all information about a process.

PCB Format

• Process ID	• Accounting Info
• Process State	• I/O Status Info
• Program Counter	• List of Open Files
• CPU Registers	• Parent Process ID
• CPU Scheduling Info	• Priority
• Memory Management Info	

Q.3 Differentiate between Ready Queue and Device Queue. (5 Marks) [PYQ ALERT: 2017]

Ready Queue	Device Queue
Contains processes ready to run.	Contains processes waiting for I/O device.
Managed by CPU scheduler.	Managed by device driver.
When CPU is free, process is taken from ready queue.	When I/O completes, process moves to ready queue.

6. SYSTEM CALLS

Q.1 Define System Call. (2 Marks) [PYQ ALERT: 2016, 2021]

Ans. System call is the interface through which a user program requests service from OS. It is generally written in C/C++.

Q.2 Types of System Calls. (5 Marks) [PYQ ALERT: 2018, 2022]

- Ans.
1. Process Control - fork(), exec(), exit(), wait()
 2. File Management - open(), read(), write(), close(), unlink()
 3. Device Management - read(), write(), ioctl()
 4. Information Maintenance - time(), getpid()
 5. Communication - pipe(), shmget(), semget()

Q.3 Write brief note on fork() and exec(). (5 Marks) [PYQ ALERT: 2019]

Ans. fork() - Creates a new process (child). Returns 0 to child and child PID to parent.
exec() - Replaces the current process image with a new program. It does not return if successful.



Common Scheduling Terms

- WT - Waiting Time.
- TAT - Turnaround Time
- CT - Completion Time
- RT - Response Time
- BT - Burst Time (CPU Time)

7. MULTIPROCESSOR SYSTEMS

Q.1 What is Multiprocessor System? (2 Marks) [PYQ ALERT: 2019, 2022]

Ans. A system having more than one CPU is called multiprocessor system.

Q.2 Types of Multiprocessor System. (2 Marks) [PYQ ALERT: 2018]

- Ans.
1. Asymmetric Multiprocessing (AMP)
 2. Symmetric Multiprocessing (SMP)

Q.3 Advantage of Multiprocessor System. (2 Marks) [PYQ ALERT: 2019]

- Ans.
1. Increased throughput
 2. Economy of scale
- (Any other valid)

8. CPU SCHEDULING

Q.1 Define CPU Scheduling. (2 Marks) [PYQ ALERT: 2014, 2016]

Ans. CPU scheduling is the process of selecting a process from ready queue and allocating CPU to it.

Q.2 What are scheduling criteria? (5 Marks) [PYQ ALERT: 2016, 2018, 2022]

- Ans.
1. CPU Utilization - Maximize
 2. Throughput - Maximize
 3. Turnaround Time - Minimize
 4. Waiting Time - Minimize
 5. Response Time - Minimize
 6. Fairness - Ensure fairness

Q.3 Explain FCFS scheduling with example. (12 Marks) [PYQ ALERT: 2014, 2022]

Ans. FCFS (First Come First Serve) - Non-preemptive. The process that comes first in ready queue gets CPU first.

Example :

Process	P1	P2	P3	P4
Burst Time	10	5	8	4

Gantt Chart :

P1	P2	P3	P4
0	10	15	23
			27

Completion Time (CT): P1=10, P2=15, P3=23, P4=27

Turnaround Time (TAT) = CT - AT (AT=0 for all)

P1=10, P2=15, P3=23, P4=27

Waiting Time (WT) = TAT - BT

P1=0, P2=10, P3=15, P4=23

Average WT = (0+10+15+23)/4 = 12

Average TAT = (10+15+23+27)/4 = 18.75

Other Scheduling Algorithms (List)

- SJF (Shortest Job First)
- SRTF (Shortest Remaining Time First)
- Priority Scheduling
- Round Robin (Time Quantum = q)
- Multilevel Queue
- Multilevel Feedback Queue

PYQ HINTS (UNIT-1)

- Definition type questions every year (Q1 short).
- CPU Scheduling long questions repeat almost every year.
- FCFS, SJF, RR numericals are high scoring.
- Scheduling criteria direct questions asked many times.
- Storage structures short question near-constant.
- Time-sharing / Multiprogramming come in short questions.

IMPORTANT FORMULAS

- WT = TAT - BT
- TAT = CT - AT
- CT = WT + BT
- Average WT = $\sum WT / n$
- Average TAT = $\sum TAT / n$
- CPU Utilization = $\frac{\text{Busy Time}}{\text{Total Time}} \times 100\%$

REMEMBER

"In OS, concept samajhna hai, ratna nahi."
Practice PYQs daily.
Draw diagrams.
Write steps in numericals.

UNIT-2 : PROCESS MANAGEMENT & SYNCHRONIZATION

1. PROCESS CONCEPT

Q.1 What is Process? (2 Marks)

[PYQ ALERT : 2014, 2016, 2017]

Ans. A process is a program in execution. It is a dynamic entity which includes program code, data, stack, heap and current activity.

Q.2 List any four characteristics of a process. (2 Marks)

[PYQ ALERT : 2018]

- Ans.
1. Dynamic
 2. Requires resources
 3. Independent unit of scheduling
 4. Has a lifetime

2. PROCESS CONTROL BLOCK (PCB)

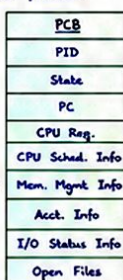
Q.1 What is PCB? List its contents. (5 Marks)

[PYQ ALERT : 2016, 2019]

Ans. PCB is a data structure used by OS to store information about each process.

Contents of PCB :

- Process ID
- Process State
- Program Counter
- CPU Registers
- CPU Scheduling Information
- Memory Management Information
- Accounting Information
- I/O Status Information
- List of Open Files



3. PROCESS STATES

Q.1 Draw and explain process state diagram. (5 Marks)

[PYQ ALERT : 2017, 2019]

Ans.



New : Process is being created.

Ready : Process is in ready queue, waiting for CPU.

Running : Process is currently executing.

Waiting : Process is waiting for I/O or event.

Terminated : Process has finished execution.

4. PROCESS SCHEDULING

Q.1 What is process scheduling? (2 Marks)

[PYQ ALERT : 2016, 2022]

Ans. Process scheduling is the activity of selecting a process from ready queue and allocating CPU to it.

Q.2 What are the objectives of scheduling? (2 Marks)

[PYQ ALERT : 2018, 2021]

- Ans.
1. High CPU utilization
 2. High throughput
 3. Low turnaround time
 4. Low waiting time
 5. Fairness

5. OPERATIONS ON PROCESSES

Q.1 List and explain operations on processes. (5 Marks)

[PYQ ALERT : 2017, 2022]

Ans. Operations :

1. Create - New process is created.
2. Terminate - Process finishes execution.
3. Suspend - Process is temporarily stopped.
4. Resume - Suspended process is restarted.
5. Block - Process waits for an event.
6. Unblock - Blocked process moved to ready queue.
7. Dispatch - CPU is allocated to process.
8. Context switch - Save state of current process and load another.

★ PYQ HINT : Operations on process comes in 5 mark questions.

6. THREADS

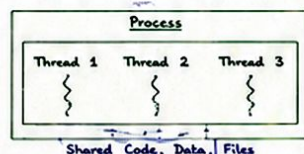
Q.1 What is Thread? List advantages. (5 Marks)

[PYQ ALERT : 2018, 2022]

Ans. Thread is the smallest unit of CPU utilization within a process.

Advantages :

1. Responsiveness
2. Resource sharing
3. Economy (less overhead)
4. Scalability
5. Better utilization of multiprocessors



7. INTER PROCESS COMMUNICATION (IPC)

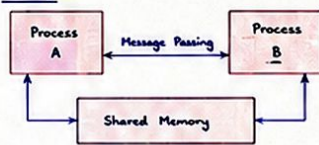
Q.1 Define IPC. List methods. (5 Marks)

[PYQ ALERT : 2019, 2022]

Ans. IPC is a mechanism that allows processes to communicate and synchronize.

Methods :

1. Shared Memory
2. Message Passing
3. Pipes
4. Sockets
5. Signals



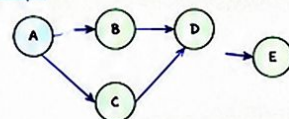
8. PRECEDENCE GRAPHS

Q.1 What is precedence graph? Draw example. (5 Marks)

[PYQ ALERT : 2019, 2022]

Ans. Precedence graph is a directed graph that represents ordering constraints between events.

Example :



Rules :

1. Nodes represent events.
2. Directed edge A → B means A must occur before B.

9. CRITICAL SECTION PROBLEM

Q.1 Define Critical Section. Write requirements. (5 Marks)

[PYQ ALERT : 2016, 2018, 2022]

Ans. Critical Section is a part of code where a process accesses shared data.

Requirements :

1. Mutual Exclusion - Only one process in critical section at a time.
2. Progress - If no process in CS, selection cannot be postponed.
3. Bounded Waiting - Every process gets chance in finite time.

★ PYQ HINT : Requirements are very important.

10. SEMAPHORES

Q.1 Define Semaphore. Explain types. (5 Marks)

[PYQ ALERT : 2016, 2019, 2022]

Ans. Semaphore is an integer variable used for process synchronization.

Types :

1. Binary Semaphore (0 or 1)
2. Counting Semaphore (0 to n)

Operations :

wait() / P() / down()
signal() / V() / up()

If SEMAPHORE = 0 → Process waits.

If SEMAPHORE > 0 → Process enters and decrement.

11. CLASSICAL SYNCHRONIZATION PROBLEMS

Q.1 List and explain any two problems. (5 Marks)

[PYQ ALERT : 2017, 2021, 2022]

Ans. 1. Producer-Consumer Problem

Synchronize producer and consumer using buffer.

2. Readers-Writers Problem

Allow multiple readers or single writer in shared data.

Other Problems :

- Dining Philosophers
- Sleeping Barber
- Cigarette Smokers

★ PYQ HINT : At least 2 problems asked many times.

12. DEADLOCK - BASIC CONCEPTS

Q.1 Define Deadlock. List necessary conditions. (5 Marks)

[PYQ ALERT : 2016, 2019, 2021, 2022]

Ans. Deadlock is a condition where set of processes are waiting forever for resources held by each other.

Necessary Conditions :

1. Mutual Exclusion
2. Hold and Wait
3. No Preemption
4. Circular Wait

★ PYQ HINT : Conditions are repeatedly asked.

13. DEADLOCK PREVENTION

Q.1 Write methods of deadlock prevention. (5 Marks)

[PYQ ALERT : 2019, 2022]

Ans. Methods :

1. Mutual Exclusion - Not possible to remove.
2. Hold and Wait - Allocate all resources at once.
3. No Preemption - Preempt resources if needed.
4. Circular Wait - Use resource ordering.

★ PYQ HINT : 4 methods important.

14. DEADLOCK AVOIDANCE - BANKER'S ALGORITHM

Q.1 Explain Banker's Algorithm. (12 Marks)

[PYQ ALERT : 2017, 2019, 2022, 2023]

Ans. Banker's Algorithm is a deadlock avoidance algorithm.

Data Structures :

1. Available[n] - Available resources
2. Max[m][n] - Maximum demand
3. Allocation[m][n] - Allocated instances
4. Need[m][n] - Max - Allocation

Steps :

1. Work = Available
2. Find i such that Need[i] ≤ Work
3. Work = Work + Allocation[i]
4. Repeat until all processes finish. If safe sequence exists → State is safe. Else → Unsafe.

Example (3 Processes, 3 Resources)

	Allocation			Max			Need		
	P ₀	P ₁	P ₂	P ₀	P ₁	P ₂	P ₀	P ₁	P ₂
R ₁	0	1	0	7	5	3	7	4	3
R ₂	2	0	0	3	2	2	1	2	2
R ₃	3	0	2	9	2	2	6	2	0

Available = [3 3 2]

Safe Sequence = < P₁, P₀, P₂ >

★ PYQ HINT : Banker's Algorithm asked most in long questions.

15. DEADLOCK DETECTION & RECOVERY

Q.1 Write methods of deadlock detection and recovery. (5 Marks)

[PYQ ALERT : 2021]

Detection :

- Resource Allocation Graph
- Wait-for Graph

Recovery :

1. Process termination
2. Resource preemption

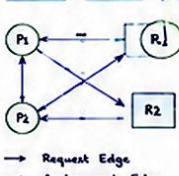
★ PYQ HINT : Short question.

COMMON DIAGRAMS

Process State Diagram



Resource Allocation Graph



Banker's Algorithm Flow



VERY SHORT QUESTIONS

1. Define Process.
2. What is PCB?
3. What is Thread?
4. Define Semaphore.
5. What is Critical Section?
6. Conditions of Deadlock?
7. What is IPC?
8. What is Precedence Graph?

IMPORTANT FORMULAS / TERMS

- Need = Max - Allocation
- Safe Sequence → No Deadlock
- wait() → Decrement and Check
- signal() → Increment
- Deadlock occurs when all 4 conditions hold.

PYQ MUST STUDY TOPICS (UNIT-2)

- ★ Deadlock (Conditions, Prevention, Avoidance, Banker's Algorithm)
- ★ Critical Section & Semaphores

- ★ Classical Problems
- ★ IPC
- ★ Process States & PCB

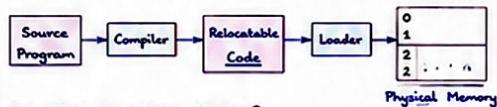
MEMORY TRICK

"Closed doors create problems, but proper synchronization opens the way."

UNIT-2 PYQ SUMMARY

Long Questions (12 Marks) → 2016, 2017, 2019, 2022, 2023
Short Questions (2/5 Marks) → Almost every year

1. ADDRESS BINDING

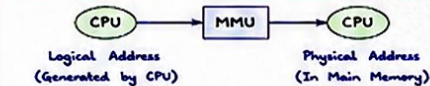


Q.1 What is address binding?

Ans. Mapping of logical (virtual) address generated by CPU to physical address in main memory is called address binding.

PYQ ALERT : Asked 2018 (2M)

2. LOGICAL VS PHYSICAL ADDRESS SPACE



Q.1 Differentiate between logical and physical address. (5 Marks)

Logical Address	Physical Address
1. Generated by CPU	1. Actual address in Main Memory
2. Part of Virtual Address Space	2. Part of Physical Address Space
3. Translated by MMU	3. Used by Memory Unit

PYQ ALERT : Asked 2016, 2019 (5M)

7. SEGMENTATION

Q.1 What is segmentation?

Ans. Logical memory is divided into variable size blocks called segments based on program structure (code, data, stack etc.).



PYQ ALERT : Asked 2017, 2022 (5M)

10. PAGE REPLACEMENT ALGORITHMS

Q.1 List and explain page replacement algorithms. (5 Marks)

Ans. When a page fault occurs and no free frame is available, a page must be replaced.

Algorithms :

1. FIFO (First In First Out)
2. LRU (Least Recently Used)
3. Optimal (MIN)

PYQ ALERT : Asked 2018, 2019, 2021, 2022, 2023 (5M)

LRU Example : (Reference String : 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3)

Step	7	0	1	2	0	3	0	4	2	0	3
Frame1	7	7	7	2	2	2	0	0	0	0	0
Frame2	-	0	0	0	0	3	4	4	4	4	4
Frame3	-	-	1	1	1	1	1	2	2	2	3
Hit/Fault	F	F	F	F	H	F	F	F	H	H	H

Total Page Faults = 8

PYQ ALERT : Asked 2023 (Numerical 5M)

13. DEMAND SEGMENTATION

Q.1 What is demand segmentation?

Ans. Segments are brought into memory only when they are needed.

PYQ ALERT : Asked 2018 (2M)

16. TRUSTED SYSTEM

Q.1 What is trusted system?

Ans. A system that behaves as expected and provides security, reliability and integrity of resources.

Features :

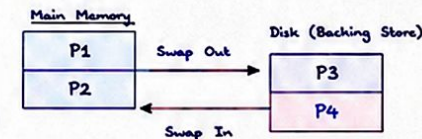
1. Correctness
2. Reliability
3. Security
4. Availability

PYQ ALERT : Asked 2022 (2M)

3. SWAPPING

Q.1 What is swapping?

Ans. Swapping is a memory management technique in which entire process is moved from main memory to secondary storage (disk) and brought back when needed.



Purpose :

1. Frees memory
2. Increases degree of multiprogramming

PYQ ALERT : Asked 2019 (5M)

4. CONTIGUOUS MEMORY ALLOCATION

Q.1 What is contiguous allocation?

Ans. Each process is allocated a single continuous block of memory.

Types :

1. Single Partition
2. Multiple Partition (Fixed / Variable)

PYQ ALERT : Asked 2016 (5M)

8. VIRTUAL MEMORY

Q.1 What is virtual memory?

Ans. It is a technique that allows execution of processes that may not be completely in main memory.

Advantage :

1. Appears to have a large memory
2. More processes can be executed
3. Better memory utilization

PYQ ALERT : Asked 2019, 2021 (5M)

IMPORTANT NUMERICALS (UNIT-3)

1. Page Replacement (FIFO, LRU, Optimal)
2. Page Faults Counting
3. Thrashing - Short Question
4. Frame Allocation Problems

PYQ ALERT : Numerical asked in 2018, 2019, 2021, 2022, 2023

5. NON-CONTIGUOUS MEMORY ALLOCATION

Q.1 What is non-contiguous allocation?

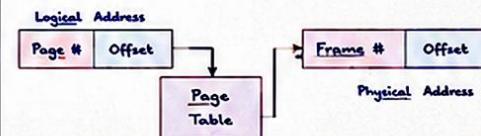
Ans. A process is allocated memory in non-contiguous blocks. It includes techniques like Paging, Segmentation and Segmentation with Paging.

PYQ ALERT : Asked 2018 (2M)

6. PAGING

Q.1 What is paging?

Ans. Logical memory is divided into fixed size blocks called pages and physical memory is divided into blocks called frames.



PYQ ALERT : Asked 2016, 2019, 2021 (5M)

9. DEMAND PAGING

Q.1 What is demand paging?

Ans. Pages are brought into memory only when they are needed (on demand).

When a page is needed but not in memory, a page fault occurs.

PYQ ALERT : Asked 2019, 2021, 2023 (5M)

12. THRASHING

Q.1 What is thrashing?

Ans. When system spends more time in paging (page faults) than in executing processes, it is called thrashing.

Causes :

1. High degree of multiprogramming
2. Not enough frames
3. Poor page replacement

Control :

1. Reduce degree of multiprogramming
2. Increase frames
3. Better page replacement

PYQ ALERT : Asked 2021 (5M)

15. SECURITY THREATS

1. Intruder (Unauthorized Access)

Q.1 What is intruder?

Ans. A person/program who tries to gain unauthorized access to system.

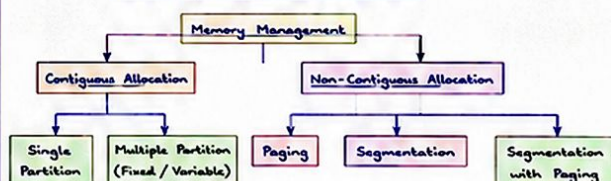
2. Computer Virus

Q.1 What is computer virus?

Ans. A program that can replicate itself and infect other programs/system.

PYQ ALERT : Asked 2017, 2019 (2M)

MEMORY MANAGEMENT SUMMARY



VERY SHORT QUESTIONS (2 MARKS)

1. Define Swapping.
2. What is Virtual Memory?
3. Define Thrashing.
4. Define Logical Address.
5. What is Demand Paging?

5 MARK QUESTIONS (IMPORTANT)

1. Explain Paging with diagram.
2. Explain Segmentation.
3. Explain Page Replacement Algorithms.
4. Explain Frame Allocation.
5. Explain Protection in OS.

LONG QUESTIONS (12 MARKS)

1. Explain Virtual Memory system in detail.
2. Explain Paging and Page Replacement with example.
3. Explain Segmentation with diagram and example.
4. Explain Thrashing and its control.
5. Explain any one security threat in detail.

MEMORY TRICK

"Page aaya tab frame mila, nahi mila to fault hua."
"More fault, less work = Thrashing"
Don't give OS a thrashing! 😊

BCA-205 OPERATING SYSTEM & UNIX

UNIT-4 : OPEN SOURCE, FILE SYSTEM & LINUX COMMANDS

1. OPEN SOURCE SOFTWARE

Q.1 What is Open Source Software? Write its advantages and disadvantages. (5 Marks) [PYQ ALERT : 2017, 2019]

Ans. Open Source Software is software whose source code is freely available to use, modify, and distribute.

Advantages :

1. Cost effective
2. Freedom to modify
3. Reliable and secure (bug fixing by community)
4. No vendor lock-in
5. Encourages innovation

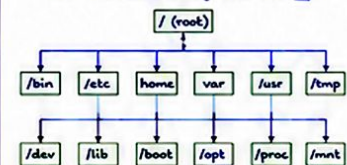
Disadvantages :

1. No guarantee / warranty
2. Requires technical knowledge
3. Support may be limited
4. Compatibility issues sometimes

2. FILE-SYSTEM HIERARCHY (LINUX)

Q.1 Draw and explain the Linux file system hierarchy. (5 Marks) [PYQ ALERT : 2018]

Ans. Linux files are organized in a hierarchical tree structure starting from root (/).



Directory	Purpose
/	Root directory of entire file system
/bin	Essential user command binaries
/etc	System configuration files
/home	Home directories of users
/var	Variable files (logs, mail, spool)
/usr	User programs and data
/dev	Device files
/tmp	Temporary files
/boot	Boot loader files
/proc	Process and kernel info (virtual)
/lib	Essential system libraries
/opt	Optional / third party software
/mnt	Used for mounting file systems

3. LOGGING IN (LINUX)

Q.1 Write steps to login in Linux system. (2 Marks) [PYQ ALERT : 2016]

Ans.

1. Start the system.
2. At login prompt, enter username.
3. Enter password.
4. After successful login, shell prompt appears.
5. Now we can execute commands.

4. TYPES OF USERS

Q.1 What are the types of users in Linux? (2 Marks) [PYQ ALERT : 2019]

Ans.

1. Super User (root) - has all privileges.
2. Normal User - limited privileges.
3. System User - created by system for running services.

5. BASIC STRUCTURE OF A COMMAND

command [options] [arguments]

Example : ls -l /home
ls → command, -l → option, /home → argument

6. IMPORTANT LINUX COMMANDS (MOST ASKED)

Command	Use / Function	Example
ls	List directory contents	ls -l /home
cp	Copy files or directories	cp file1 file2
mv	Move / rename files	mv old new
rm	Remove file	rm file.txt
cat	Display file contents	cat file.txt
more / less	View file page by page	more file.txt
head	Show first lines	head -n 10 file
tail	Show last lines	tail -n 10 file
touch	Create empty file	touch new.txt
mkdir	Create directory	mkdir dir1
rmdir	Remove empty directory	rmdir dir1
pwd	Print working directory	pwd
cd	Change directory	cd /etc
whoami	Show current user	whoami
ps	Show running processes	ps -ef
kill	Terminate process	kill pid
clear	Clear terminal	clear
wc	Count lines/words/chars	wc file.txt
sort	Sort lines	sort file.txt
uniq	Remove duplicate lines	uniq file.txt
grep	Search pattern in file	grep "os" file.txt
echo	Display text	echo "Hello"

7. PERMISSIONS (FILE MODES)

Q.1 Explain file permissions in Linux with example. (5 Marks) [PYQ ALERT : 2016, 2021]

Ans. Each file has 3 types of permissions for User (u), Group (g) and Others (o).

r = read (4) w = write (2) x = execute (1)

Numeric Representation :

I	rwx
7	4+2+1 (rwx)
6	4+2 (rw-)
5	4+1 (r-x)
4	4 (r--)
3	2+1 (-wx)
2	2 (-w-)
1	1 (--x)
0	0 (---)

Example : -rwxr-xr--

u	g	o
7	5	4

Numeric mode = 754

8. CHMOD, CHOWN, CHGRP

Command	Use	Example
chmod	Change file mode	chmod 755 file.txt
chown	Change owner	chown user file.txt
chgrp	Change group	chgrp group file.txt

Examples :

- chmod 755 file.txt (rwx r-x r-x)
- chmod 644 file.txt (rw- r-- r--)
- chown rahman file.txt
- chgrp students file.txt

9. HARD LINK & SOFT LINK

Q.1 Differentiate between hard link and soft link. (5 Marks) [PYQ ALERT : 2018]

Point	Hard Link	Soft Link
Definition	Another name for same file	Pointer to another file
Inode	Same inode	Different inode
Cross FS	Not possible	Possible
Original deleted	Link still works	Link broken
Command	ln file link	ln -s file link

10. I/O REDIRECTION

Q.1 What is I/O redirection? Give examples. (5 Marks) [PYQ ALERT : 2017, 2022]

Ans. I/O redirection is used to change the normal input or output. Data is taken from or sent to a file instead of terminal.

Types :

1. Output Redirection (>)
2. Input Redirection (<)
3. Append Output (>>)

Examples :

ls > list.txt (output to file)
cat < list.txt (input from file)
echo "hello" >> file.txt (append)
grep "os" < file.txt > out.txt

11. PIPES (|)

Q.1 What are pipes in Linux? Give example. (3 Marks) [PYQ ALERT : 2019]

Ans. Pipe (|) is used to send output of one command as input to another command.

Example :

ls -l | more
ps -ef | grep bash
cat file.txt | sort | uniq

12. ENVIRONMENT VARIABLES

Q.1 What are environment variables? Give example of some variables. (5 Marks) [PYQ ALERT : 2016, 2021]

Ans. Environment variables are dynamic values that affect the behavior of processes.

Common Variables :

Variable	Purpose
HOME	Home directory
PATH	Search path for commands
USER	Current username
SHELL	Current shell
PWD	Current working directory
LANG	Language setting
TZ	Time zone
DISPLAY	Display for GUI
TMP	Temporary directory

13. DISPLAY / SET / UNSET

Command	Use	Example
echo \$VAR	Display value	echo \$PATH
export VAR=val	Set variable	export PATH=/bin
set	Display all variables	set
unset VAR	Remove variable	unset PATH

Example :

export NAME=AKU
echo \$NAME (Output : AKU)
unset NAME

14. WHICH, WHEREIS, WHATIS, TYPE

Command	Use	Example
which	Shows path of command	which ls
whereis	Shows binary, source & man	whereis gcc
whatls	Short description	whatls mkdir
type	Tells type of command	type cd

15. USERS RELATED COMMANDS

Command	Use	Example
who	Show logged in users	who
whoami	Show current user	whoami
id	Show user id and groups	id
finger	Show user info	finger user
w	Show who is logged in	w

16. PROCESS RELATED COMMANDS

Command	Use	Example
ps	Show process status	ps -ef
top	Dynamic process view	top
kill	Send signal to process	kill 1234
pkill	Kill by name	pkill firefox

17. DISK USAGE COMMANDS

Command	Use	Example
df	Disk free space	df -h
du	Directory usage	du -sh dir
free	Memory usage	free -m

18. FILE COMPARISON COMMANDS

Command	Use	Example
diff	Compare two files	diff f1 f2
cmp	Compare byte by byte	cmp f1 f2
comm	Compare sorted files	comm f1 f2

19. JOB CONTROL COMMANDS

Command	Use	Example
&	Run command in background	sleep 100 &
fg	Bring background job to foreground	fg %1
bg	Resume job in background	bg %1
jobs	List jobs	jobs

20. IMPORTANT SHORTCUT KEYS

Ctrl + C → Terminates current command
Ctrl + Z → Suspend current command
Ctrl + D → Logout / Exit shell
Ctrl + L → Clear screen
Ctrl + A → Move cursor to start of line
Ctrl + E → Move cursor to end of line
Ctrl + U → Cut from start to cursor
Ctrl + K → Cut from cursor to end

21. COMMON FILE TYPES

Extension	Type
.txt	Text file
.sh	Shell script
.c	C source file
.cpp	C++ source file
.o	Object file
.exe	Executable file
.doc/pdf/jpg	Document / Image

22. COMMAND CATEGORY SUMMARY

1. File/Directory Mgmt : ls, cp, mv, rm, mkdir
2. File View : cat, more, less, head, tail
3. Search & Filter : grep, cut, sort, uniq, wc
4. Process Mgmt : ps, top, kill, pkill
5. Disk & Memory : df, du, free
6. User Info : who, whoami, id, finger
7. System Info : uname, history, clear
8. Permissions : chmod, chown, chgrp

PYQ HINTS (UNIT-4)

- ★ Command based questions come every year.
- ★ Permissions (chmod, chown, links) are very frequent.
- ★ Redirection, Pipes & Environment variables are important short questions.
- ★ File-system hierarchy and login steps asked many times.

1. VI EDITOR (INTRODUCTION)

Q.1 What is vi editor? Write its features.
(2 Marks) [PYQ ALERT : 2016, 2018]

Ans. vi is a screen oriented text editor available in UNIX/Linux.

Features :

- Fast and efficient
- Three modes - Command, Insert, Last line
- Powerful editing commands
- Supports search and replace
- Can recover files (.exrc, .swp)
- Works in terminals

2. MODES OF VI EDITOR

Q.1 Explain three modes of vi editor.

(5 Marks) [PYQ ALERT : 2014, 2016, 2018]

Mode	Purpose	Prompt	How to Enter	How to Exit
Command Mode	Used to give commands	-	vi filename (or Esc)	i, a, o to go to Insert mode : to Last line mode
Insert Mode	Used to insert/edit text	- INSERT -	i (insert), a (append), o (open)	Esc (to Command mode)
Last Line Mode	Used for file operations	:	: (colon)	Enter Esc (to Command mode)

PYQ HINT : Modes are short questions every year.

3. BASIC VI COMMANDS (COMMAND MODE)

Q.1 Write any ten vi commands with their use.

(5 Marks) [PYQ ALERT : 2016, 2017, 2018, 2022]

Command	Use
i	- Insert before cursor
a	- Append after cursor
o	- Open new line below cursor
dd	- Delete current line
yy	- Yank (copy) current line
p	- Put (paste) after cursor
u	- Undo last change
/text	- Search forward text
n	- Repeat last search
:w	- Write (save) file
:q	- Quit vi
:wq	- Write and quit
:q!	- Quit without saving

PYQ HINT : vi commands are very frequent.

4. YANK, PUT & DELETE

Q.1 Explain yank, put and delete with example.
(5 Marks) [PYQ ALERT : 2016, 2018]

Ans.

- Yank (yy) - Copy line(s) or text.
Example : yy copies current line.
- Put (p) - Paste yanked text after cursor.
Example : p pastes the yanked line.
- Delete (dd) - Delete line or text.
Example : dd deletes current line.

PYQ HINT : Short but scoring question.

5. SEARCH & REPLACE

Q.1 How to search and replace text in vi?
(5 Marks) [PYQ ALERT : 2017, 2019]

Ans.

Search : /text → Search forward
?text → Search backward
Press n → Next match

Replace : :s/old/new/ → Replace first
:s/old/new/g → Replace all in line
:%s/old/new/g → Replace in whole file

PYQ HINT : Search & replace asked many times.

6. .EXRC FILE

Q.1 What is .exrc file? Give example.
(2 Marks) [PYQ ALERT : 2018, 2022]

Ans.

.exrc is a startup file executed whenever vi starts.
It contains vi commands.

Example (.exrc file content) :

```
set nu
set ai
set wrap
set ic
```

PYQ HINT : Very short question.

7. VI COMMANDS SUMMARY TABLE

Type	Commands	Use
Cursor Movement	h, j, k, l	Left, Down, Up, Right
Word Movement	w, b, e	Next word, Prev word, End word
Line Movement	O, \$, ^	Start line, End line, First non-blank
Delete	x, dd, d\$	Char delete, Line delete, Till end
Copy (Yank)	yy, y\$	Line yank, Till end yank
Paste	p, P	After cursor, Before cursor
Undo/Redo	u, Ctrl+r	Undo, Redo
Save/Quit	:w, :q, :wq, :q!	Write, Quit, Write&Quit, Quit!

PYQ HINT : Commands list asked many times.

8. SHELL (INTRODUCTION)

Q.1 What is Shell? Write its types.
(2 Marks) [PYQ ALERT : 2014, 2016, 2018, 2019]

Ans. Shell is a command interpreter that acts as an interface between user and OS.

Types of Shell :

- Bourne Shell (sh)
- C Shell (csh)
- Korn Shell (ksh)
- Bash Shell (bash)

PYQ HINT : Shell types asked repeatedly.

9. REDIRECTION IN SHELL

Q.1 Explain input and output redirection.
(5 Marks) [PYQ ALERT : 2017, 2021, 2022]

Ans. Redirection is used to change the default input/output.

Type	Symbol	Use	Example
Output	>	Overwrite output	ls > list.txt
Redirection	>>	Append output	ls >> list.txt
Input	<	Take input from file	sort < file.txt
Error Redirection	2>	Redirect errors	sort 2> err.txt
Both	&>	Redirect all output	sort &> all.txt

PYQ HINT : Redirection short & long both.

10. PIPES (|) AND FILTERS

Q.1 What is pipe? Write example.
(3 Marks) [PYQ ALERT : 2016, 2018, 2021]

Ans. Pipe (|) is used to send output of one command as input to another.

Example :

```
ls -l | more
cat file.txt | grep "abc" | wc -l
```

Common Filters :

grep, sort, uniq, wc, cut, head, tail, tr

PYQ HINT : Pipe questions are frequent.

11. ALIASES

Q.1 What is alias? Give example.
(2 Marks) [PYQ ALERT : 2019, 2021]

Ans. Alias is a user defined short name for a long command.

Example :

```
alias ll="ls -l"
alias cls="clear"
```

Unalias : unalias ll

List aliases : alias

PYQ HINT : Short question.

12. JOB CONTROL

Q.1 Write job control commands with use.
(5 Marks) [PYQ ALERT : 2017, 2019, 2022]

Command	Use
&	- Run command in background
jobs	- List background jobs
fg %n	- Bring job n to foreground
bg %n	- Resume job n in background
Ctrl+Z	- Suspend current job
kill %n	- Kill job n

PYQ HINT : Job control long question.

13. AWK (PATTERN SCANNING AND PROCESSING)

Q.1 What is awk? Write basic example.
(5 Marks) [PYQ ALERT : 2018, 2021, 2023]

Ans. awk is a powerful programming language used for pattern scanning and processing.

Example :

```
awk '{print $1,$3}' file.txt (Print 1st and 3rd field)
awk -F: '{print $1}' /etc/passwd (Print first field using : as separator)
```

PYQ HINT : awk asked in long questions.

14. LINUX FEATURES

Q.1 Write important features of Linux.
(5 Marks) [PYQ ALERT : 2016, 2019, 2021, 2022, 2025]

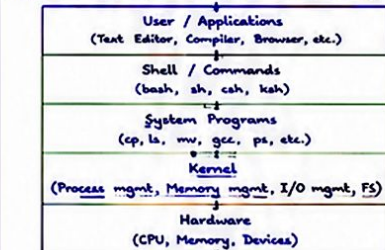
Ans.

- Open Source and free
- Multi-user and multitasking
- Portable (runs on many platforms)
- Strong security and stability
- Supports networking
- Powerful command line interface
- Hierarchical file system
- Customizable and flexible

PYQ HINT : Features asked many times.

15. LINUX SYSTEM ARCHITECTURE (COMPONENTS)

Q.1 Draw and explain components of Linux system.



Ans. Components :

- Hardware - Actual physical devices.
- Kernel - Core part, manages resources.
- System Programs - Utility programs.
- Shell - Interface between user and kernel.
- Applications - User programs.

PYQ HINT : Architecture diagram asked in many exams.

16. PROCESS, THREAD & FILE MANAGEMENT

Q.1 Write role of Linux in process, thread and file management.
(5 Marks) [PYQ ALERT : 2019, 2022, 2023]

Ans.

- Process Management - Creates, schedules, terminates processes.
Commands : ps, top, kill, nice
- Thread Management - Supports multi-threading.
Commands : ps -L, top -H
- File Management - Create, delete, read, write files.
Commands : ls, cp, mv, rm, cat, chmod

PYQ HINT : Important long question.

17. DEVICE DRIVERS

Q.1 What is device driver? Give example.
(2 Marks) [PYQ ALERT : 2018, 2021]

Ans. Device driver is a program that allows OS to communicate with hardware device.
Examples : Printer driver, Audio driver, Network driver, USB driver.

PYQ HINT : Short question.

IMPORTANT VI / SHELL SHORTCUTS

VI SHORTCUT KEYS

i	Insert before cursor
a	Append after cursor
o	Open new line
Esc	Command mode
:w	Save file
:q	Quit
:wq	Save and quit
:q!	Quit without save

SHELL SHORTCUT KEYS

Ctrl + C	Terminate
Ctrl + Z	Suspend
Ctrl + D	Logout/Exit
Ctrl + L	Clear screen
Ctrl + R	Search history
!!	Repeat last command
Tab	Auto complete

PYQ HINTS (UNIT-5)

- vi editor commands, modes asked every year.
- Shell types, redirection and pipes very frequent.
- Job control and awk asked many times.
- Linux features, components asked repeatedly.
- Short + Long questions from this unit are high scoring.



1. LOGIN TO LINUX SYSTEM

Q1 How do we login to Linux system?

(2 Marks) [PYQ ALERT : 2016, 2019]

Ans. Steps to login :

1. At login prompt, enter username.
2. Press Enter.
3. Type password.
4. After successful login, shell prompt appears.

Example :

```
ubuntu login: zoha
Password: *****
zoha@ubuntu:~$
```

2. USERNAMES & GROUPS

Q1 What are users and groups in Linux?

(2 Marks) [PYQ ALERT : 2018, 2022]

Ans. Every user has a unique username and UID.

- Groups are collection of users.
- Users in same group share common permissions.
- Primary group and supplementary groups exist.

Commands :

Command	Use
id	Show user id and groups
groups	Show groups of current user
whoami	Show current username
groupadd	Create new group (sudo groupadd g1)
useradd	Create user (sudo useradd u1)
usermod	Modify user (sudo usermod -G g1 u1)

3. PASSWORD MANAGEMENT

Q1 How to set and change password?

(2 Marks) [PYQ ALERT : 2016, 2018, 2021]

Ans. passwd command is used.

- User can change own password.
- Root can change any user's password.

Commands :

Command	Use
passwd	Change own password
passwd username	Change other user's password
passwd -e username	Expire password (force change)

[PYQ ALERT] : Password related questions are frequently asked.

4. LINUX DOCUMENTATION

Q1 Write any two sources of Linux documentation.

(2 Marks) [PYQ ALERT : 2017, 2022]

Ans. Sources :

1. man pages (manual pages)
2. info pages
3. --help option in commands
4. Documentation in /usr/share/doc

Example : man ls (opens manual of ls command)
ls --help (short help)

5. PATHNAMES

Q1 Differentiate between absolute and relative pathname.

(5 Marks) [PYQ ALERT : 2016, 2019, 2021]

Ans.

Absolute Pathname	Relative Pathname
1. Starts from root "/"	1. Starts from current directory
2. Full path is specified	2. Partial path is specified
3. Example : /home/zoha/file.txt	3. Example : docs/file.txt
4. Independent of current directory	4. Depends on current directory

6. LINUX DIRECTORIES & FILES

Q1 List important directories in Linux.

(5 Marks) [PYQ ALERT : 2018, 2021, 2023]

Ans.

Directory	Purpose
/	Root directory
/bin	Essential user commands
/etc	System configuration files
/home	Home directories of users
/usr	User programs and data
/var	Variable files (logs, mail etc.)
/tmp	Temporary files
/boot	Boot loader files
/dev	Device files
/proc	Process and kernel info (virtual)

7. LINUX INSTALLATION

Q1 What are general steps of Linux installation?

(5 Marks) [PYQ ALERT : 2019, 2022]

Ans. Steps :

1. Boot from installation media (CD/DVD/USB)
2. Select language and keyboard
3. Partition disk / use default
4. Set hostname and timezone
5. Create user and root password
6. Install software packages
7. Configure boot loader (GRUB)
8. Reboot the system

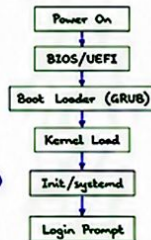
8. BOOTING PROCESS

Q1 Write booting sequence of Linux system.

(5 Marks) [PYQ ALERT : 2019, 2023]

Ans.

1. BIOS/UEFI performs POST
2. Loads boot loader (GRUB)
3. GRUB loads kernel
4. Kernel initializes hardware
5. Mounts root filesystem
6. Starts init/systemd process (PID 1)
7. System reaches login prompt

**9. USER ACCOUNTS MANAGEMENT**

Q1 Write commands to manage user accounts.

(5 Marks) [PYQ ALERT : 2016, 2018, 2021]

Ans.

Command	Use
useradd	Add new user
userdel	Delete user
usermod	Modify user
passwd	Set/Change password
chage	Change password ageing
id	Show user and group id
su	Switch user

10. FILE SYSTEMS

Q1 List common file systems in Linux.

(2 Marks) [PYQ ALERT : 2017, 2022]

Ans. Common File Systems :

1. ext4 (Extended File System)
2. ext3
3. xfs
4. btrfs
5. vfat (FAT32)
6. ntfs (Windows)

Check file system :

```
df -T (show file systems)
mount (show mounted systems)
```

11. SPECIAL FILES

Q1 What are special files? List examples.

(2 Marks) [PYQ ALERT : 2018, 2022]

Ans. Special files represent devices.

Types :

1. Character special files
2. Block special files
3. Named pipes (FIFO)
4. Sockets

Examples :

/dev/null	- Null device
/dev/tty	- Terminal
/dev/sda	- Disk device
/dev/zero	- Zero device
/dev/random	- Random generator

12. BACKUP & RESTORATION

Q1 Write commands for backup and restoration.

(5 Marks) [PYQ ALERT : 2019, 2023]

Ans.

Backup Commands :

Command	Use
cp -r source dest	Backup using copy
tar -cvf backup.tar /home/zoha	Create tar archive
tar -czvf backup.tar.gz /home/zoha	Create compressed backup

Restoration Commands :

Command	Use
cp -r backup dest	Restore using copy
tar -xvf backup.tar	Extract archive
tar -xzvf backup.tar.gz	Extract compressed archive

13. UNIX SYSTEM ADMINISTRATION (ROLES & TASKS)

Q1 Write role of UNIX system administrator.

(5 Marks) [PYQ ALERT : 2016, 2022, 2023]

Ans. Role of System Administrator :

- Manage users, groups and permissions
- Install and upgrade system software
- Configure and maintain file systems
- Monitor system performance
- Ensure security and data protection
- Schedule backups and disaster recovery
- Manage printers and device drivers
- Troubleshoot system problems

14. INSTALLATION METHOD & CLASSIFICATION

Q1 What are methods of Linux installation?

(2 Marks) [PYQ ALERT : 2017, 2021]

Ans. Methods :

1. Graphical (GUI) installation
2. Text / Console installation

Classification of Linux Distributions :

1. Debian based : Ubuntu, Linux Mint
2. Red Hat based : Fedora, CentOS, RHEL
3. SUSE based : openSUSE, SLES
4. Arch based : Arch Linux, Manjaro

15. PRE-INSTALLATION CHECKS

Q1 List pre-installation checks before installing Linux.

(2 Marks) [PYQ ALERT : 2018, 2022]

Ans.

1. Check hardware compatibility
2. Check disk space availability
3. Backup important data
4. Check BIOS settings
5. Bootable media (USB/DVD) ready

IMPORTANT COMMANDS QUICK LIST

Command	Use	Command	Use
uname -a	System information	ls -l	List details
df -h	Disk free space	cd dir	Change directory
du -sh dir	Directory usage	pwd	Print working dir
top	Running processes	mkdir file	Create empty file
kill pid	Kill process	rm file	Remove file
free -m	Memory usage	mv src dest	Move file
ps -ef	Show all processes	cat file	View file

VERY SHORT QUESTIONS (2 MARKS)

1. How to login in Linux?
2. What is root user?
3. What is /etc directory?
4. What is passwd command?
5. What is absolute pathname?
6. What is /dev/null?
7. What is backup?
8. What is boot loader?
9. What is useradd command?
10. What is file system?

5 MARK QUESTIONS (IMPORTANT)

1. Role of system administrator.
2. Steps of Linux installation.
3. Difference between absolute and relative pathname.
4. Commands for user management.
5. Write file backup and restore commands.

PYQ HINTS (UNIT-6)

- ★ Admin related questions asked regularly (2016, 2022, 2023).
- ★ Pathnames, user management, backup are high scoring.
- ★ Short + long questions repeated from this unit.

UNIT-7 : CPU SCHEDULING ALGORITHMS

1. FIRST COME FIRST SERVE (FCFS)

Q1 Explain FCFS scheduling.

(5 Marks) [PYQ ALERT : 2016, 2018, 2023]

Ans.

- FCFS is the simplest CPU scheduling algorithm.
- Processes are executed in the order in which they arrive in ready queue.
- Non-preemptive algorithm.
- Easy to implement but has high average waiting time.

Numerical Example :

Process	AT	BT	CT	TAT	WT
P1	0	24	24	24	0
P2	1	3	27	26	23
P3	2	3	30	28	25

Gantt Chart :

0	P1	24	P2	27	P3	30
---	----	----	----	----	----	----

Average Waiting Time = $(0+23+25)/3 = 16$ Average Turnaround Time = $(24+26+28)/3 = 26$

2. SHORTEST JOB FIRST (SJF) - Non Preemptive

Q1 Explain SJF scheduling with example.

(5 Marks) [PYQ ALERT : 2017, 2019, 2021]

Ans.

- Process with shortest CPU burst time is executed first.
- Non-preemptive.
- Gives minimum average waiting time.
- Requires knowing burst time in advance.

Example :

Process	AT	BT	CT	TAT	WT
P1	0	7	8	8	1
P2	1	4	12	11	7
P3	2	1	9	7	6
P4	3	4	16	13	9

Gantt Chart :

0	P1	7	P3	8	P2	12	P4	16
---	----	---	----	---	----	----	----	----

Avg WT = $(1+7+6+9)/4 = 5.75$ Avg TAT = $(8+11+7+13)/4 = 9.75$

3. SHORTEST REMAINING TIME FIRST (SRTF)

Q1 Explain SRTF with example.

(5 Marks) [PYQ ALERT : 2018, 2021, 2023]

Ans.

- Preemptive version of SJF.
- Always selects process with shortest remaining time.
- If new process arrives with smaller remaining time, current process is preempted.

Example :

Process	AT	BT
P1	0	8
P2	1	4
P3	2	9
P4	3	5

Gantt Chart :

0	P1	1	P2	4	P1	17	P3	26
---	----	---	----	---	----	----	----	----

Answers :

Process	CT	TAT	WT
P1	17	17	9
P2	5	4	0
P3	26	24	15
P4	10	7	2

Avg WT = $(9+0+15+2)/4 = 6.5$ Avg TAT = $(17+4+24+7)/4 = 13$

4. PRIORITY SCHEDULING

Q1 Explain Priority scheduling with example.

(5 Marks) [PYQ ALERT : 2016, 2019, 2022]

Ans.

- Each process is assigned a priority.
- Higher priority process executes first.
- Non-preemptive (unless specified).
- Starvation may occur for low priority processes.

Example (Lower number = Higher Priority) :

Process	AT	BT	Priority
P1	0	10	3
P2	1	1	1
P3	2	2	4
P4	3	1	2

Gantt Chart :

0	P1	10	P2	1	P4	12	P3	14
---	----	----	----	---	----	----	----	----

Note : Non-preemptive Priority

5. ROUND ROBIN (RR)

Q1 Explain Round Robin scheduling with example.

(5 Marks) [PYQ ALERT : 2017, 2020, 2023]

Ans.

- Preemptive scheduling algorithm.
- Each process gets a fixed time quantum (q).
- If process does not finish in q, it is preempted and placed at end of ready queue.

Example : Time Quantum (q) = 2

Process	AT	BT
P1	0	5
P2	1	4
P3	2	2
P4	3	1

Gantt Chart :

0	P1	2	P2	4	P3	6	P1	8	P4	9	P2	11	P1	12
---	----	---	----	---	----	---	----	---	----	---	----	----	----	----

Answers :

Process	CT	TAT	WT
P1	12	12	7
P2	11	10	6
P3	6	4	2
P4	9	6	5

Avg WT = $(7+6+2+5)/4 = 5$ Avg TAT = $(12+10+4+6)/4 = 8$

6. MULTILEVEL QUEUE SCHEDULING

Q1 Explain Multilevel Queue scheduling.

(5 Marks) [PYQ ALERT : 2016, 2018]

Ans.

- Ready queue is partitioned into multiple queues.
- Each queue has its own scheduling algorithm.
- Queues have fixed priority.
- Higher priority queue executes first.

Example :

- System Processes - RR (q = 4) - Highest Priority
- Interactive Processes - RR (q = 8)
- Batch Processes - FCFS - Lowest Priority

Note : No process can move between queues.

7. MULTILEVEL FEEDBACK QUEUE (MFQ)

Q1 Explain MFQ scheduling.

(5 Marks) [PYQ ALERT : 2017, 2021, 2023]

Ans.

- Extension of Multilevel Queue.
- Processes can move between queues.
- New processes enter highest priority queue.
- If process uses full quantum, it is moved to lower queue.
- Prevents starvation.

Example (typical) :

Q1 (q=4) → Q2 (q=8) → Q3 (FCFS)

Feedback : Lower queue → Higher queue

(if I/O bound or waits long time).

8. COMPARISON OF CPU SCHEDULING ALGORITHMS

Algorithm	Preemptive	Avg WT	Starvation	Complexity	Best Use
FCFS	No	High	No	Very Low	Simple, small systems
SJF	No	Minimum	Yes	Low	Batch systems
SRTF	Yes	Low	Yes	Medium	Time sharing
Priority	No/Yes	Low	Yes	Medium	Real time (with care)
Round Robin	Yes	Medium	No	Low	Time sharing
Multilevel Queue	No	Low	Yes	Medium	Mixed environments
MFQ	Yes	Low	No	High	General purpose OS

PYQ HINT : Comparison table asked every year.

9. EVALUATION CRITERIA

Q1 Write criteria for CPU scheduling.

(5 Marks) [PYQ ALERT : 2016, 2019, 2022]

Ans. Criteria :

- CPU Utilisation - Should be maximum.
- Throughput - Number of processes completed per unit time.
- Turnaround Time - Should be minimum.
- Waiting Time - Should be minimum.
- Response Time - Should be minimum.
- Fairness - Each process gets fair share.

10. REAL TIME SCHEDULING

Q1 Explain Real Time Scheduling.

(5 Marks) [PYQ ALERT : 2018, 2023, 2025]

Ans.

- Real-time systems have strict timing constraints.
- Tasks must complete within deadline.

Types :

- Hard Real-Time - Missing deadline is failure.
- Soft Real-Time - Missing deadline reduces performance.

Algorithms :

- Rate Monotonic (RM)
- Earliest Deadline First (EDF)

Used in : Embedded systems, Robotics, Control systems.

11. I/O ORGANIZATION

Q1 Write I/O organisation in OS.

(5 Marks) [PYQ ALERT : 2016, 2017]

Ans.

- OS manages communication between CPU, memory and I/O devices.
- Controlless manage device operations.
- Two methods :
 - Programmed I/O (Polling)
 - Interrupt Driven I/O
- Uses device drivers to operate hardware.

12. I/O DEVICES

Q1 Classify I/O devices.

(5 Marks) [PYQ ALERT : 2019, 2022]

Ans. Types of I/O Devices :

- Block Devices - e.g., Disk, CD/DVD
- Character Devices - e.g., Keyboard, Printer
- Network Devices - e.g., Network Card, Modem
- Storage Devices - e.g., HDD, SSD, Pen Drive
- Communication Devices - e.g., NIC, Modem
- Special Purpose devices - e.g., Scanner, Camera

13. I/O BUFFERING

Q1 What is buffering? Explain types.

(5 Marks) [PYQ ALERT : 2016, 2021, 2023]

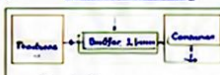
Ans.

- Buffering is a technique to store data temporarily before transfer to device.
- Improves speed and CPU utilization.

Types of Buffering :

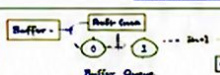
- Single Buffering
- Only one bit moving Example.

2. Double Buffering



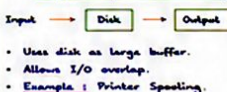
- Two buffers.
- After buffer full, producer waits.
- Simple but low efficiency.

3. Circular Buffering



- Two buffers.
- While one is being used, other is filled.
- Used in OS queues.

4. Spooling (Simultaneous Peripheral Operations On-Line)



- Uses disk as large buffer.
- Allows I/O overlap.
- Example : Printer Spooling.

IMPORTANT FORMULAS (Scheduling)

- Completion Time (CT)
 - Turnaround Time (TAT) = CT - AT
 - Waiting Time (WT) = TAT - BT
 - Response Time (RT) = First Response - AT
- Where,
AT = Arrival Time
BT = Burst Time

VERY SHORT QUESTIONS (2 MARKS)

- Explain FCFS.
- What is Time Quantum?
- What is Starvation?
- What is Preemptive Scheduling?
- Give example of Block Device.

5 MARK QUESTIONS (IMPORTANT)

- Explain SJF with example.
- Explain Round Robin with example.
- Write Priority Scheduling.
- Explain Multilevel Queue.
- Write any five criteria.

LONG QUESTIONS (12 MARKS)

- Compare all CPU scheduling algorithms.
- Explain MFQ with diagram.
- Explain Real Time Scheduling.
- Explain I/O Organisation and Buffering.
- Solve any scheduling numerical.

PYQ HINTS (UNIT-7)

- Scheduling numericals asked every year.
- Comparison questions very frequent.
- Round Robin, SRTF most important.
- Real Time, I/O topics repeatedly asked.

EXAM ALERT

- Long Q = 12 Marks
- 5 Marks Q = 5 Marks
- 2 Marks Q = 2 Marks
- Diagram = Steps = High Score

BCA-205 OPERATING SYSTEM & UNIX**MICRO CHEATSHEET - ALL UNITS (1 to 7)**

EXAM DATE : 16 SEPT 2026

TIME : 10 AM - 1 PM

FULL MARKS : 60

PYQ YEARS (VERIFIED)

2014, 2016, 2017, 2018,
2019, 2021, 2022, 2023,
2025 (2024 not supplied)

UNIT 1 : INTRODUCTION, CPU SCHEDULING & I/O**1. OS INTRODUCTION**

Q1 What is OS? Write its functions. (5 Marks) [PYQ 2016, 2018, 2023]

Ans: OS is system software that acts as interface between user and hardware.

Functions / Services :

- Process Management
- Memory Management
- File Management
- I/O Management
- Security & Protection
- Resource Allocation
- Error Detection

2. TYPES OF OS

Q1 List types of OS. (2 Marks) [PYQ 2017, 2021]

- Batch OS
- Time Sharing OS
- Real Time OS
- Distributed OS
- Network OS

4. CPU SCHEDULING ALGORITHMS (Summary)

Algorithm	Preemptive	Avg WT	Throughput	Best Use
FCFS	No	High	Low	Simple systems
SJF	No	Minimum	High	Batch systems
SRTF	Yes	Low	High	Time sharing
Priority	Yes/No	Low	Medium	Real time
Round Robin	Yes	Medium	Medium	Time sharing
Multilevel Qnq	No	Low	Medium	Mixed systems
Multilevel Feedback	Yes	Low	High	General purpose

5. SCHEDULING CRITERIA

Q1 Write criteria for CPU scheduling. (5 Marks) [PYQ 2016, 2019, 2023]

Ans:

- CPU Utilisation - Maximize
- Throughput - Maximize
- Turnaround Time - Minimize
- Waiting Time - Minimize
- Response Time - Minimize
- Fairness - Avoid starvation

6. REAL TIME SCHEDULING

Q1 What is real time scheduling? (2 Marks) [PYQ 2018, 2021]

Ans: Scheduling tasks with strict timing constraints.

Types :

- Hard Real Time - Missing deadline is failure.
- Soft Real Time - Missing deadline reduces performance.

7. I/O ORGANIZATION & DEVICES

Q1 Classify I/O devices. (2 Marks) [PYQ 2014, 2021]

Ans:

- Block Devices - Disk, SSD, CD/DVD
- Character Devices - Keyboard, Printer, Terminal
- Network Devices - NIC, Modem
- Storage Devices - HDD, SSD, Pen Drive
- Communication - Modem, NIC
- Special Purpose - Camera, Sensor

7. I/O ORGANIZATION & DEVICES

Q1 Classify I/O devices. (2 Marks) [PYQ 2014, 2021]

Ans:

- Block Devices - Disk, SSD, CD/DVD
- Character Devices - Keyboard, Printer, Terminal
- Network Devices - NIC, Modem
- Storage Devices - HDD, SSD, Pen Drive

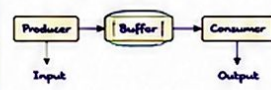
8. I/O BUFFERING

Q1 What is buffering? Types? (2 Marks) [PYQ 2016, 2021]

Ans: Buffering is temporary storage to improve I/O speed.

Types :

- Single Buffering
- Double Buffering
- Circular Buffering
- Spooling (Simultaneous Peripheral Operations On-Line)

**PYQ HINTS (UNIT-1)**

- OS Introduction very frequent.
- CPU Scheduling & criteria most repeated.
- System Calls asked many times.
- I/O Organization, Buffering important short & long.

UNIT 2 : PROCESS MANAGEMENT & SYNCHRONIZATION**1. PROCESS CONCEPT**

Q1 What is a process? (2 Marks) [PYQ 2016, 2018]

Ans: Program in execution.

Process elements in PCB :

- PID
- State
- PC
- Registers
- Priority
- Memory info
- I/O info

2. PROCESS STATES

Q1 Draw and explain process states. (5 Marks) [PYQ 2016, 2019]

**3. THREADS**

Q1 What is thread? (2 Marks) [PYQ 2017, 2021]

Ans: Lightweight process.

Multiple threads of a process share same resources.

Types :

- User Level Threads
- Kernel Level Threads
- Hybrid Threads

4. IPC (INTER PROCESS COMMUNICATION)

Q1 List IPC methods. (2 Marks) [PYQ 2018, 2022]

Ans:

- Shared Memory
- Message Passing
 - Direct Communication
 - Indirect Communication

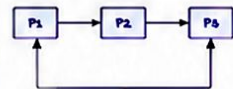
5. PRECEDENCE GRAPHS

Q1 What is precedence graph? (2 Marks) [PYQ 2019, 2021]

Ans: Directed graph showing order of resource allocation.

No cycle -> Safe

Cycle -> Deadlock

**5. CRITICAL SECTION**

Q1 What is critical section? (2 Marks) [PYQ 2016, 2018]

Ans: Code segment that accesses shared resources.

7. SEMAPHORES

Q1 What is semaphore? (2 Marks) [PYQ 2017, 2021]

Ans: Integer variable used for synchronization.

Types :

- Binary Semaphore (0,1)
- Counting Semaphore (0...n)

Operations :

wait (S) / P(S)
signal (S) / V(S)

8. DEADLOCK

Q1 What is deadlock? (2 Marks) [PYQ 2016, 2018]

Ans: Set of processes waiting forever for resources.

9. DEADLOCK CONDITIONS

Q1 List conditions. (2 Marks) [PYQ 2016, 2018]

Ans:

- Mutual Exclusion
- Hold and Wait
- No Preemption
- Circular Wait

10. DEADLOCK HANDLING

Q1 List methods. (5 Marks) [PYQ 2017, 2022]

Ans:

- Prevention
- Avoidance (Banker's Algo)
- Detection and Recovery

PYQ HINTS (UNIT-2)

- Process states, PCB asked repeatedly.
- Semaphores & Classical problems important.
- Deadlock: Conditions, Prevention, Banker's Algo very important.

UNIT 3 : MEMORY MANAGEMENT & SECURITY**1. ADDRESS BINDING**

Q1 Differentiate logical & physical address. (2 Marks) [PYQ 2016, 2019]

Logical Address	Physical Address
Generated by CPU	Actual address in main memory

2. SWAPPING

Q1 What is swapping? (2 Marks) [PYQ 2016, 2021]

Ans: Moving process between main memory and secondary storage.

3. CONTIGUOUS ALLOCATION

Q1 What is contiguous allocation? (2 Marks) [PYQ 2019, 2019]

Ans: A process is allocated continuous memory blocks.

- Types :
- Single Partition
 - Multiple Partition

4. PAGING

Q1 Define paging. (2 Marks) [PYQ 2016, 2018]

Ans: Memory is divided into fixed size pages and frames.

5. SEGMENTATION

Q1 Define segmentation. (2 Marks) [PYQ 2017, 2019]

Ans: Memory is divided into variable size segments.

Seg#	Base	Limit
1	1000	400
2	2000	700
3	4000	600

6. VIRTUAL MEMORY

Q1 What is virtual memory? (2 Marks) [PYQ 2016, 2021]

Ans: Technique that allows execution of large processes using secondary storage.

7. DEMAND PAGING

Q1 What is demand paging? (2 Marks) [PYQ 2016, 2019]

Ans: Load page in memory only when needed.

8. PAGE REPLACEMENT ALGORITHMS

Q1 List algorithms. (2 Marks) [PYQ 2016, 2018, 2022]

- FIFO
- LRU
- Optimal

PYQ HINTS (UNIT-3)

- Paging, Segmentation, Virtual Memory most asked.
- Page Replacement numericals every year.
- Security short questions important.

UNIT 4 : OPEN SOURCE, FILE SYSTEM & LINUX COMMANDS**1. OPEN SOURCE**

Q1 What is Open Source? (2 Marks) [PYQ 2019, 2019]

Ans: Software whose source code is freely available to use, modify and distribute.

2. FILE SYSTEM HIERARCHY

Q1 Draw Linux file hierarchy. (2 Marks) [PYQ 2016, 2018]

**3. IMPORTANT LINUX COMMANDS (Quick List)**

Command	Use	Command	Use
ls	List files	pwd	Print working dir
cp	Copy files	cd	Change directory
mv	Move/Rename	df	Disk free space
rm	Remove	du	Disk usage
cat	Concatenate	chmod	Change mode
more/less	View file	chown	Change owner
head	First lines	chgrp	Change group
tail	Last lines	wc	Word/line count
touch	Create file	sort	Sort lines
mkdir	Create dir	grep	Search pattern
rmdir	Remove dir	echo	Display text

5. REDIRECTION & PIPES

Q1 Write redirection symbols. (2 Marks) [PYQ 2019, 2019]

Ans:

- > Output to file
- >> Append output
- < Input from file
- | Pipe (send output to input)
- 2> Error redirection

5. LINKS & ENV VARIABLES

Q1 Types commands (2 Marks) [PYQ 2014, 2022]

Ans:

- Hard Link - Multiple names for same file.
- Soft Link (Symbolic) - Shortcut to file.

Environment Variable:
Used to store system settings.
Example: PATH, HOME, USER

PYQ HINTS (UNIT-5)

- Linux commands short questions frequent.
- Redirection, Pipes, Links important.
- Tab context queried many times.

UNIT 6 : LINKED UTILITIES ALGORITHMS (Numericals)**1. VI EDITOR**

Q1 Modes of vi editor. (2 Marks) [PYQ 2016, 2018]

Ans:

- Command Mode (Esc)
- Insert Mode (i)
- Last Line Mode (:)

3. SHELL

Q1 Types of shell. (2 Marks) [PYQ 2019, 2019]

Ans:

- Bourne Shell (sh)
- C Shell (csh)
- Korn Shell (ksh)
- Bash Shell (bash)

3. SHELL FEATURES

Q1 List shell features. (2 Marks) [PYQ 2016, 2021]

Ans:

- Redirection
- Pipes
- Filters
- Aliases
- Job Control
- AWK

5. LINKUP ROBIN (q=2)

Q1 Write implementation features. (2 Marks) [PYQ 2019, 2022]

Ans:

- tar -cvf backup.tar file/ P1 P2 P3 P4

Avg WT = 2 Avg TAT = 4.67

IMPORTANT NUMERICALS LIST

- FCFS, SJF, SRTF
- Priority, RR
- Banker's Algorithm
- Paging (Pages/Frames)
- Page Replacement (FIFO, LRU, Optimal)

★ Numericals carry high marks!

UNIT 7 : CPU SCHEDULING ALGORITHMS (Numericals)**A. FCFS (Example)**

Process	AT	BT	Order	Gantt Chart
P1	0	5		
P2	1	3		
P3	2	8		

WT = (0,4,6) Avg WT = 3.33
TAT = (5,7,14) Avg TAT = 8.67

B. SJF (Non Preemptive)

Process	AT	BT	Order	Gantt Chart
P1	0	6		
P2	1	2		
P3	2	8		

WT = 2 Avg WT = 2
TAT = 7.33

C. SRTF (Preemptive & P1AT)

Process	AT	BT	Order	Gantt Chart
P1	0	6		
P2	1	2		
P3	2	8		

Avg WT = 2 Avg TAT = 6

D. Round Robin (q=2)

Process	AT	BT	Order	Gantt Chart
P1	0	5		
P2	1	3		
P3	2	8		

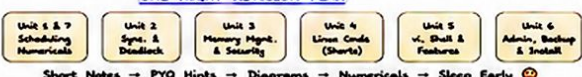
Avg WT = 2 Avg TAT = 4.67

IMPORTANT NUMERICALS LIST

- FCFS, SJF, SRTF
- Priority, RR
- Banker's Algorithm
- Paging (Pages/Frames)
- Page Replacement (FIFO, LRU, Optimal)

TOP 10 EXAM TIPS

- Draw diagrams neatly.
- Write steps & formulas.
- Show all calculations.
- Underline final answers.
- Manage time (3 hrs).
- Attempt long questions first.
- Use headings & points.
- Write clean & readable.
- Don't leave numericals.
- Revise PYQ hints.

ONE NIGHT REVISION PLAN

Short Notes -> PYQ Hints -> Diagrams -> Numericals -> Sleep Early

MORNING CHECKLIST

- All formulas & diagrams revised
- Scheduling Gantt practice
- Commands list memorized
- PYQ Hints read once
- Stay calm & positive

ALL THE BEST!
YOU CAN DO IT!

UNIT 1 : INTRODUCTION, CPU SCHEDULING & I/O

1. OS Introduction

OS is system software that acts as an interface between user & hardware.

Services : Program Execution, I/O Operation, File Management, Memory Mgmt, Protection, Communication, Error Detection.

PYQ: 2016, 2018 (SM)

2. Types of OS

Batch, Multiprogramming, Time-sharing, Real-time, Distributed, Network, Embedded.

PYQ: 2016, 2019 (SM)

3. Multiprogramming vs Time-sharing

Multiprogramming	Time-sharing
CPU switches when process waits for I/O	CPU switches after short time quantum
Goal: Max CPU Utilization	Goal: Min Response Time
Non-interactive	Interactive

PYQ: 2017, 2021 (SM)

4. CPU Scheduling Criteria

Criteria	Definition
CPU Utilization	Maximize
Throughput	Maximize
Turnaround Time	Minimize
Waiting Time	Minimize
Response Time	Minimize
Fairness	Ensure fairness

PYQ: 2016, 2018, 2021 (2/SM)

5. I/O Organization

- I/O Devices, Device Drivers, DMA
- I/O Modules: User I/O, Kernel I/O
- I/O buffering improves performance.

PYQ: 2018, 2022 (SM)

6. Real-time Scheduling

Hard Real-time: Missing deadline = failure.
 Soft Real-time: Missing reduces performance.
 Algorithms: Rate Monotonic (RM), EDF.

PYQ: 2018, 2023 (SM)

7. CPU Scheduling Algorithms

Algorithm	Preemptive	Best Use	PYQ
FCFS	No	Simple, small systems	2016,18
SJF	No	Batch systems	2018,19
SRTF	Yes	Time sharing	2018,21
Priority	Yes/No	Real-time systems	2016,21
Round Robin	Yes	Time sharing	2016,22
Multilevel Queue	No	System with queues	2018
Multilevel Feedback Queue	Yes	Mixed environments	2019,23

8. Short Notes (2 Marks)

System Calls, Storage Structures, Multiprocessor Systems, Multiple Processor Scheduling, I/O Devices, Buffering.

PYQ: Repeated every year.

9. Important Numericals

- FCFS, SJF, SRTF, Priority, RR, MQ, MFQ
- Avg Waiting Time, Avg Turnaround Time
- Gantt Chart, Ready Queue

Short Answer

- What is OS?
- Types of OS.
- Criteria of CPU Scheduling.
- Real-time vs Time-sharing.
- I/O buffering.

PYQ: 2014-2023 (Repeated)

UNIT 1 MEMORY TRICK

"FCFS Pehle Aaya Pehle Paya,"
 "SJF Chhota Kaam Pehle Aaya,"
 "SRTF Time Bacha Sabko Bhaya."



UNIT 2 : PROCESS MANAGEMENT & SYNCHRONIZATION

1. Process Concept

Program in execution is called process.
 Process Control Block (PCB) stores info.

PYQ: 2016, 2018 (SM)

PCB Contains
PID, State, PC, Registers, CPU Sched Info, Memory Info, I/O Info, Accounting Info

2. Process States



PYQ: 2016, 2017, 2019 (SM)

3. Threads

- Lightweight process.
- Share code, data, files.
- Each thread has own stack, PC, registers.

PYQ: 2018, 2021 (SM)

4. IPC (Inter Process Communication)

Methods: Shared Memory, Message Passing, Pipe, Message Queue, Socket.

PYQ: 2017, 2019 (SM)

5. Precedence Graph

Directed acyclic graph shows ordering of process of processes.

PYQ: 2016, 2021 (2M)

6. Critical Section

Part of program that accesses shared data.
 Problems: Mutual Exclusion, Race Condition.

PYQ: 2016, 2018 (SM)

7. Semaphores

Semaphore is an integer variable used for synchronization.

Types: Binary, Counting

Operations: wait(), signal()

PYQ: 2017, 2019, 2021 (SM)

8. Classical Problems

Producer-Consumer, Readers-Writers, Dining Philosophers.

PYQ: 2018, 2022 (SM)

9. Deadlock

Deadlock: Process waits forever for resources.

PYQ: 2016, 2018 (2/SM)

10. Deadlock Prevention

Mutual Exclusion, Hold & Wait, No Preemption, Circular Wait - break any one.

PYQ: 2019, 2021 (SM)

11. Deadlock Avoidance

Safe State, Unsafe State

Banker's Algorithm.

PYQ: 2016, 2022, 2023 (SM)

12. Deadlock Recovery

Process Termination

Resource Preemption

Backtracking

PYQ: 2018 (SM)

Important Diagram

PCB	Semaphore Operations
- OS - PCB - Registers - CPU Sched Info - Mem Info - I/O Info - Accounting Info	<pre>wait(S): while (S <= 0) : S = S - 1 signal(S): S = S + 1</pre>

Banker's Algorithm Steps

- Available Allocation → Available.
- Need, based Allocation the allocations.
- Find safe seq, one turn then end state.
- If of available 2 need - If Available 20 signal. with a available two of donation.
- If all finish → Safe.

SHORT NOTES (2 Marks)

PCB, Threads, IPC, Critical Section, Semaphore, Deadlock.

PYQ: Asked every year.

UNIT 3 : MEMORY MANAGEMENT & SECURITY

1. Memory Address

- Logical Address: Generated by CPU
- Physical Address: Actual in memory
- Address binding: Compile, Load, Execution.

PYQ: 2016, 2018 (2M)

2. Swapping

Move process from memory to disk & bring back.

PYQ: 2019 (2M)

3. Contiguous Allocation

Single, Fixed Partition, Variable Partition.

PYQ: 2016, 2018 (SM)

4. Non-contiguous Allocation

Paging, Segmentation.

PYQ: 2016 (2M)

5. Paging

- Logical memory → Pages
- Physical memory → Frames
- Page Table used for mapping.

PYQ: 2016, 2018, 2022 (SM)

Page#	Frame#
0	2
1	5
2	1
3	3
...	...
n	m

6. Segmentation

Variable size segments, Base & Limit.

PYQ: 2019 (2M)

7. Virtual Memory

- Extends memory using disk.
- Demand Paging allows loading on demand.

PYQ: 2017, 2021 (2M)

8. Page Replacement Algorithms

Algorithm	Idea	PYQ
FIFO	Oldest	2016
LRU	Least Recently Used	2018
Optimal	Replace farthest in future	2019

9. Frame Allocation

Equal, Proportional, Priority, Local, Global.

PYQ: 2021 (2M)

10. Thrashing

Too many page faults → System slow.

PYQ: 2018 (2M)

11. Protection

- Base & Limit, ACL, Capability List.

PYQ: 2016, 2019 (2M)

12. Security

Unauthorized access prevention.

PYQ: 2018 (2M)

13. Intruders

Vulnerabilities exploited.

PYQ: 2019 (2M)

14. Computer Viruses

Self-replicating programs.

PYQ: 2017 (2M)

15. Trusted System

Trusted OS ensures security.

PYQ: 2020 (2M)

NUMERICALS (Unit 3)

- Page Replacement.
- Demand Paging
- Frame Allocation
- Paging Table
- Segmentation
- Thrashing Short Q.

Memory Trick

"Page table dhoondo jaldi,
 "Frame chuno sahi sahi,
 "Thrashing se bacho bhai,
 LRU rakho yaad bhai!"



UNIT 4 : OPEN SOURCE, FILE SYSTEM & LINUX COMMANDS

1. Open Source

Free to use, modify, distribute.
 Examples: Linux, Android.

PYQ: 2016 (2M)

2. File System Hierarchy

- /root
- /bin, /home, /etc,
- /usr
- /tmp

3. Important Commands (Short Notes)

- ls, cp, mv, rm
- cat, more, less, head, tail
- touch, mkdir, rmdir, find
- cd, pwd, cp, mv, rm

PYQ: Repeated every year.

4. Permissions (chmod)

rwx, w, x, -
 Example: 755 = rwxr-xr-x

5. Links

- Hard Link (Same inode)
- Soft Link (Different inode)
- Hard Link (different in future)

PYQ: 2016, 2018, 2021 (SM)

6. I/O Redirection & Pipes

> (output), < (input), >> (append), | (pipe)

PYQ: 2018 (2M)

7. Environment Variables

PATH, HOME, USER, export, set, echo \$VAR

PYQ: 2017 (2M)

VERY SHORT Q (2 MARKS)

- Open Source.
- File System Hierarchy
- chmod commands.
- Hard vs. Relative Link.
- Linux Redirection
- Environment Variable.
- Explain any 5 commands.
- Booting Process.

UNIT 5 : VI EDITOR, SHELL & LINUX FEATURES

1. vi Editor Modes

Command Mode	Insert Mode
Insert Mode	i/o
Last Line Mode	:

2. vi Important Commands

- :w, :q, :wq, :x, :e, :r, :s, :p, :d, :y, :u, :c, :u, :r, :p, :d, :y, :u, :c

3. Shell

Command Interpreter
 Types: sh, csh, bash, ksh
 PYQ: 2016 (2M)

4. Shell Features

- Redirection, Pipes, Filters
- Aliases, Job Control
- AWK - Pattern scanning

PYQ: 2017, 2019 (SM)

5. Linux Features

- Multiuser, Multitasking
- Portable, Secure
- Open Source, Reliable

PYQ: 2018 (2M)

Micro Memory Trick

- vi ke teen mode yaad,
- Esc se command mode,
- i se insert mode shuru,
- ing se file ko karo soon!



UNIT 6 : LINUX UTILITIES & UNIX ADMINISTRATION

1. Login & Users

- username, password, whoami, id, groups

PYQ: 2016 (2M)

2. Linux Installation

- Steps: Boot, Partition, Install, Configure, User Create

PYQ: 2018 (SM)

3. Files and File Systems

- ext4, xfs, vfat
- mount, umount
- df, du, fsck

PYQ: 2017 (2M)

4. Backup & Restoration

- tar, cp, rsync
- Backup, Restore files

PYQ: 2019 (SM)

SHORT Q (2 MARKS)

- Login steps
- User/Group mgmt
- Password change
- Absolute vs Relative Path
- Installation steps
- Filesystems in Linux
- Backup command
- Booting Process

ALL THE BEST !

Stay Calm, Stay Confident, Do Smart Study Success is Yours!

